Subspaces

Geometric Algorithms

CAS CS 132

Practice Problem

$$A = \begin{bmatrix} 1 & 0 & -1 & 2 \\ 2 & 0 & 2 & 3 \\ 1 & 4 & 0 & 2 \end{bmatrix} \xrightarrow{\substack{R_1 \leftarrow R_1 + R_2 \\ R_3 \leftarrow 2R_3 \\ R_2 \leftrightarrow R_3}} \begin{bmatrix} 3 & 0 & 1 & 5 \\ 2 & 8 & 0 & 4 \\ 2 & 0 & 2 & 3 \end{bmatrix} = B$$

Consider the following pair of matrices A *and* B *which are row equivalent. Find a matrix* E *such that* $EA = B$

Answer

$$R_1 \leftarrow R_1 + R_2$$
$$R_3 \leftarrow 2R_3$$
$$R_2 \leftrightarrow R_3$$

$$A = \begin{bmatrix} 1 & 0 & -1 & 2 \\ 2 & 0 & 2 & 3 \\ 1 & 4 & 0 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 3 & 0 & 1 & 5 \\ 2 & 8 & 0 & 4 \\ 2 & 0 & 2 & 3 \end{bmatrix} = B$$

Objectives

- Introduce the fundamental notions of **subspaces** and **bases**
- Extend our intuitions about planes in $\mathbb{R}^3$ to subspaces in $\mathbb{R}^n$
- Connected subspaces to matrices so that we can use the techniques we been honing in this course

Keywords

subspace

closed under addition

closed under scaling

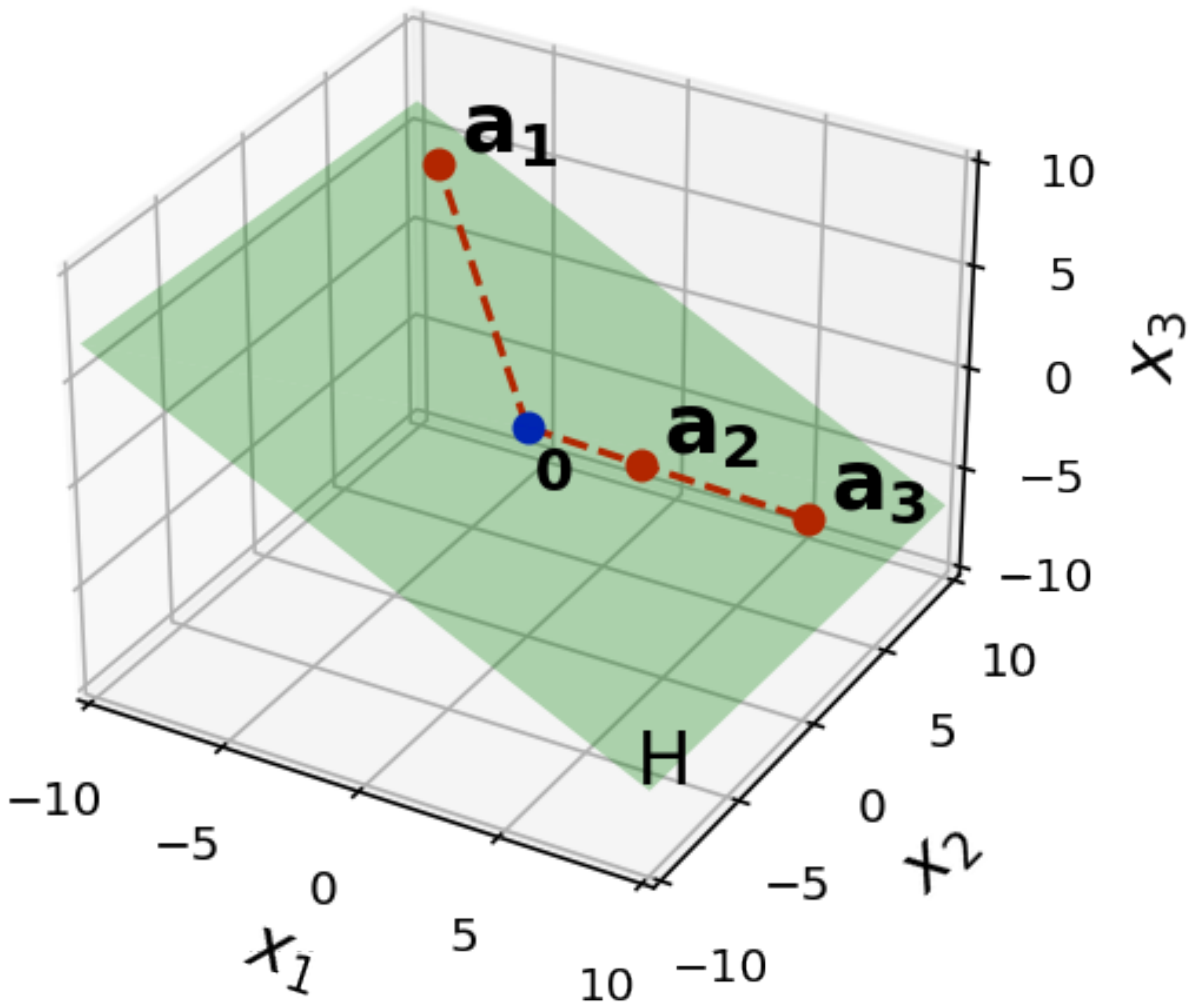
column space

null space

basis

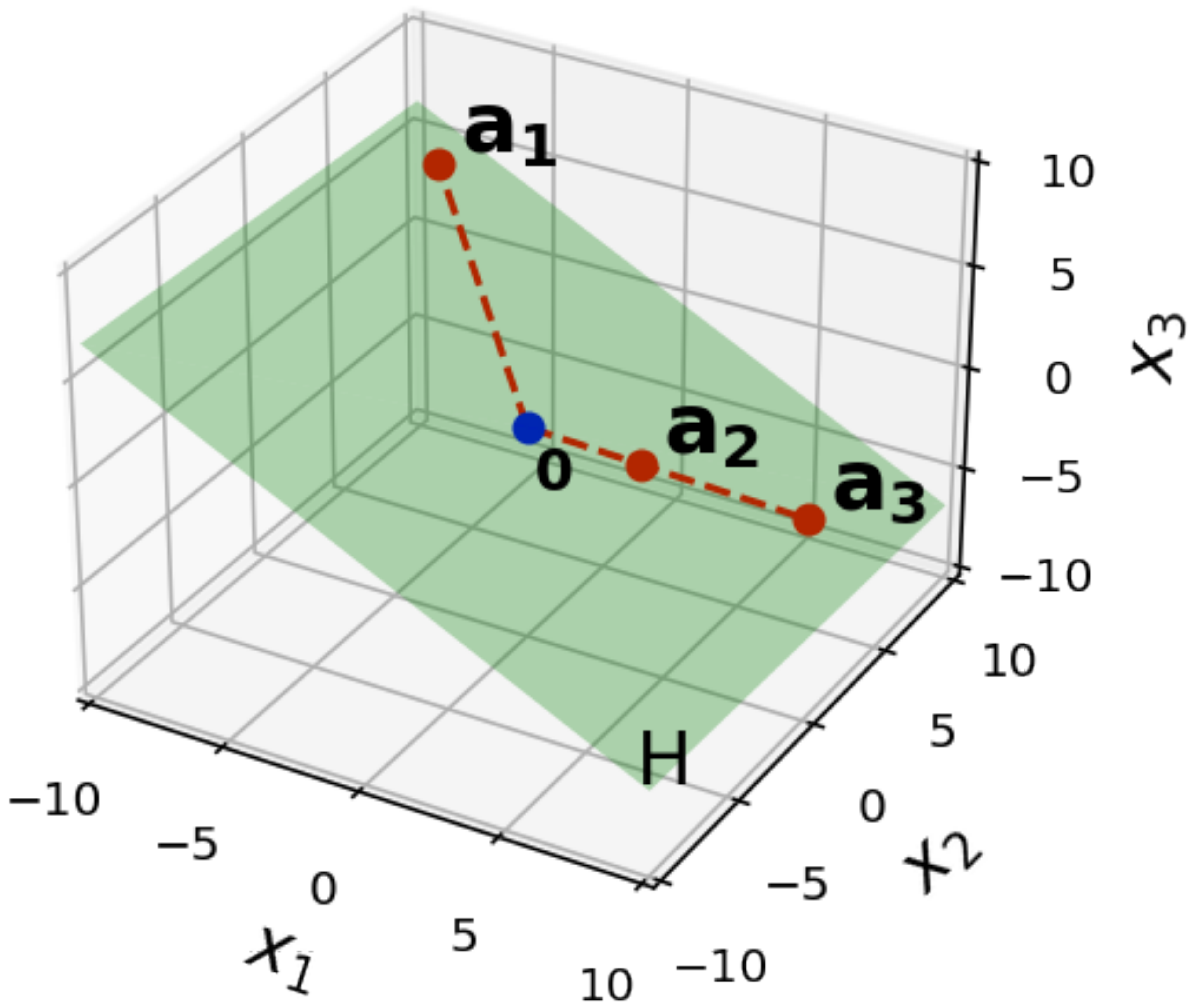
Subspaces

Subspaces



Subspaces

"sub" means "part of" or "below"



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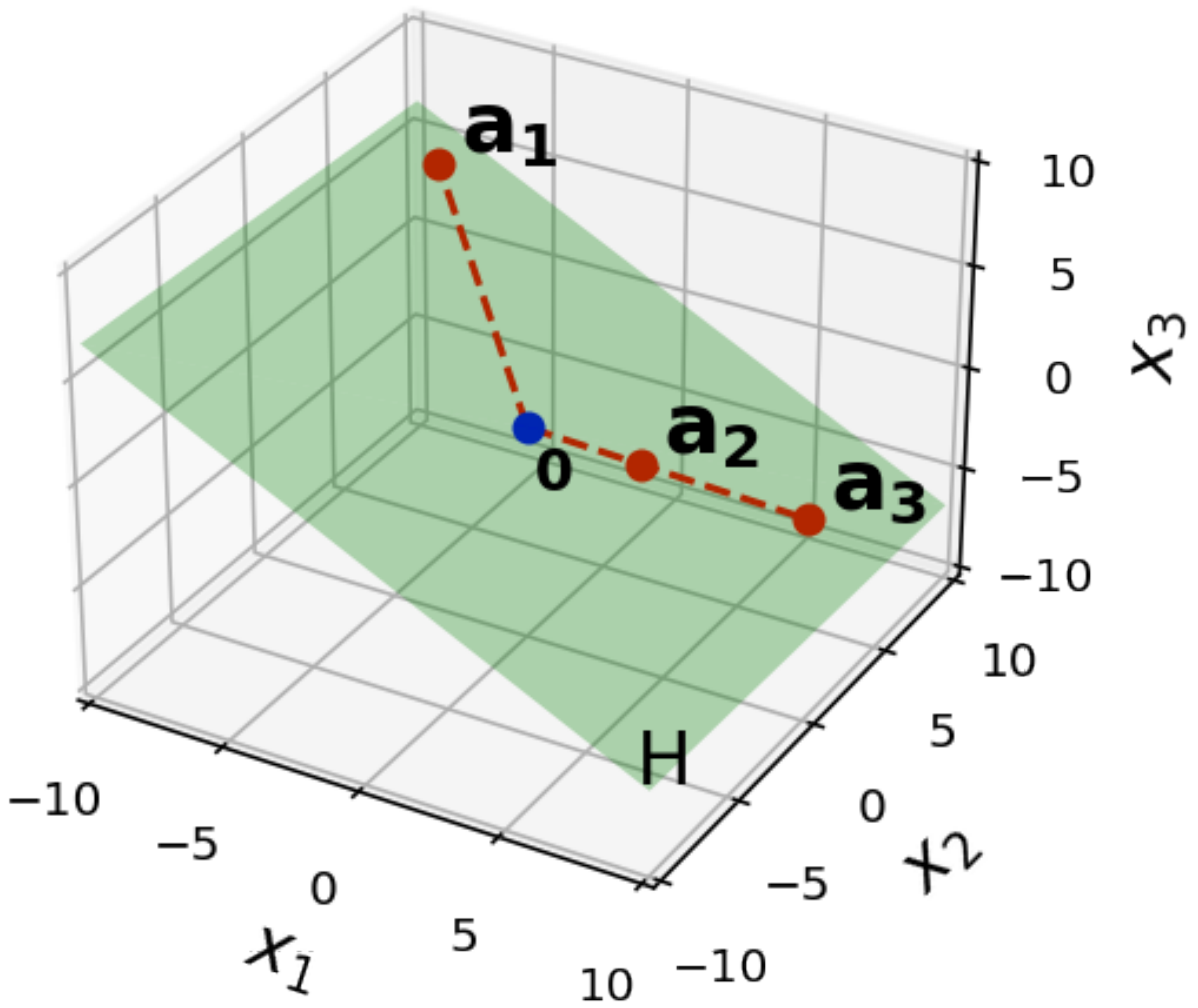
A plane in $\mathbb{R}^3$ looks like a tilted copy of $\mathbb{R}^2$



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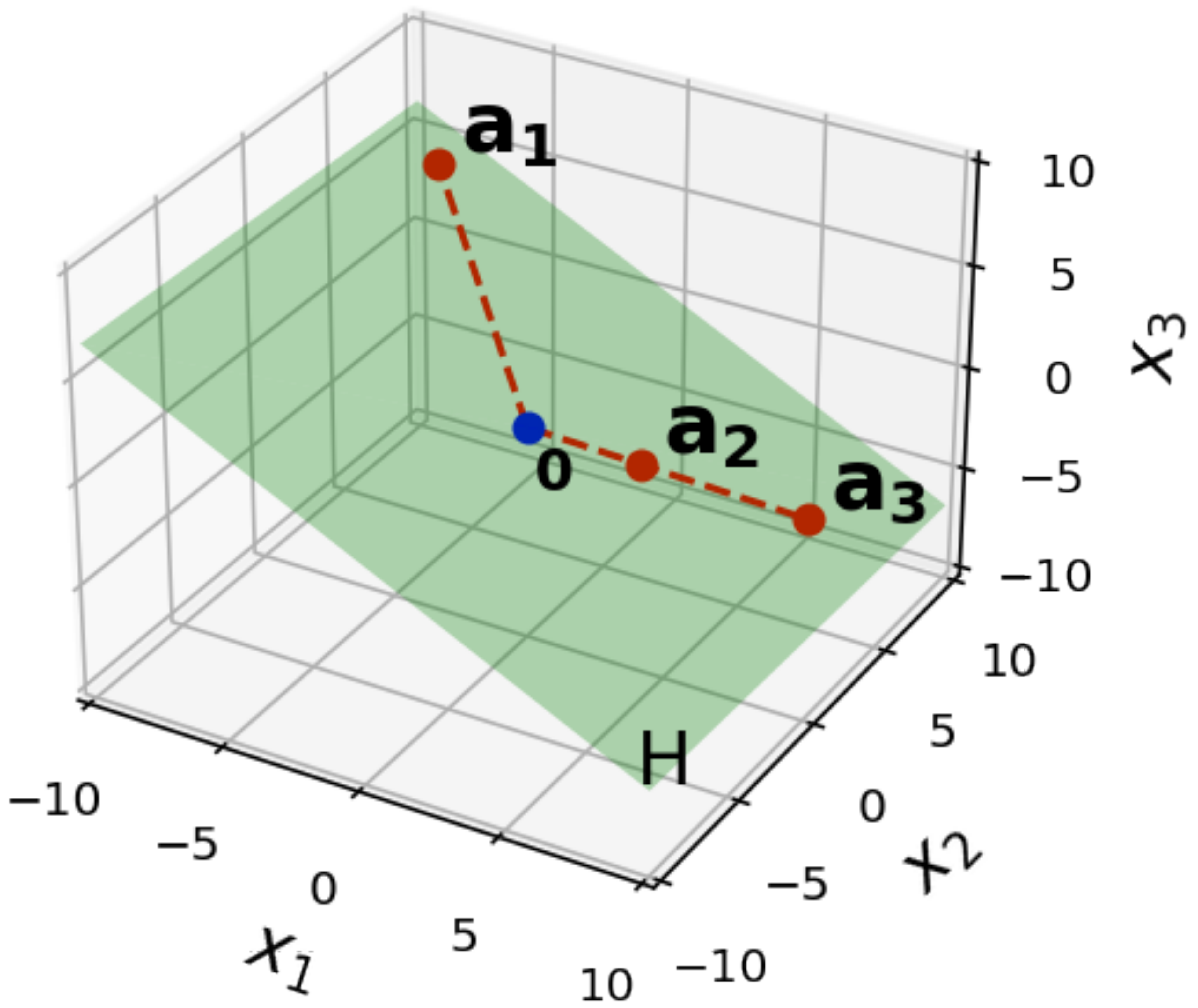
Subspaces *generalize* this



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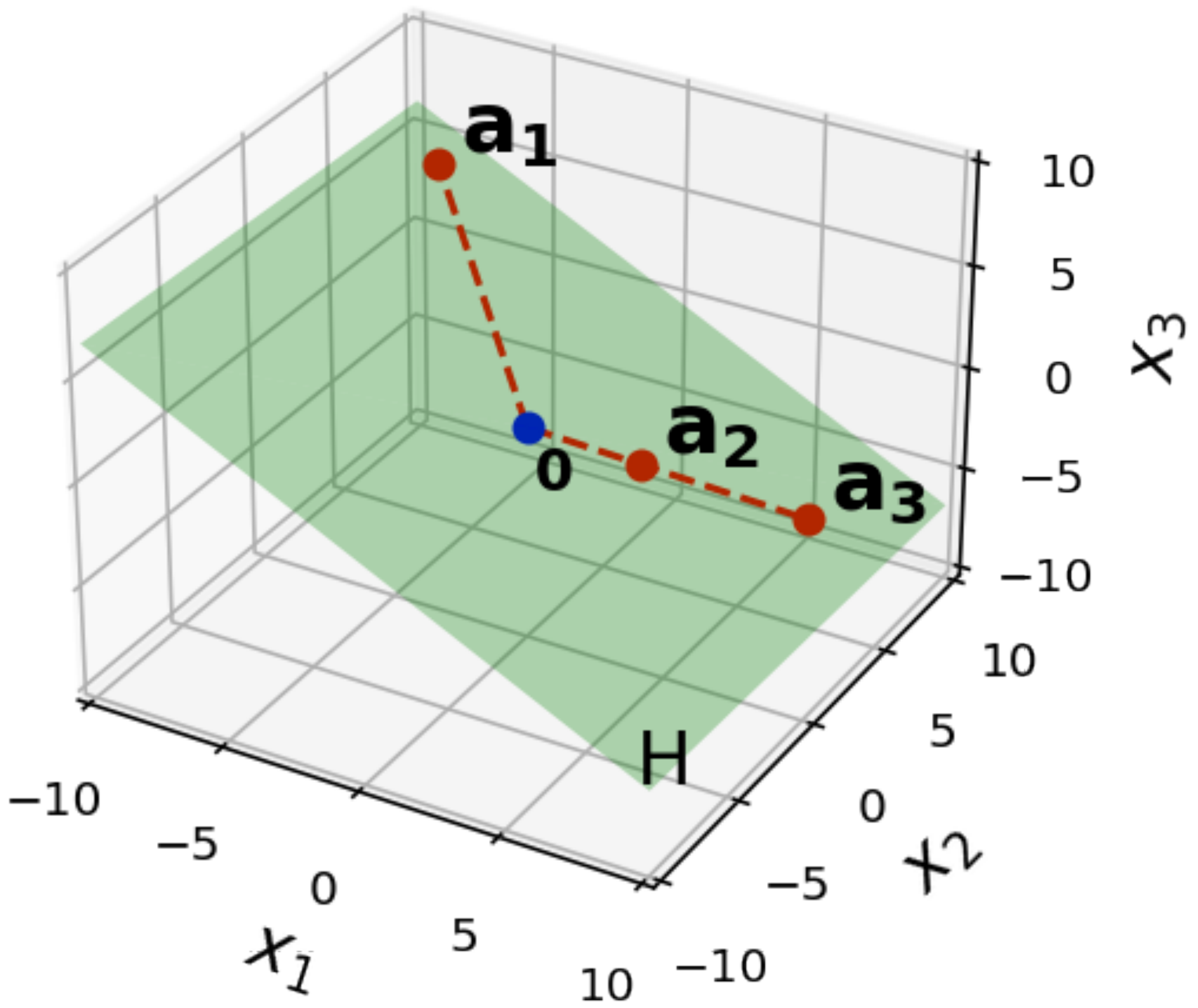
Subspaces *generalize* this

Ex. We can look at "tilted copies" of $\mathbb{R}^7$ sitting in $\mathbb{R}^{11}$



An Aside: Flatland, Relativity, Higher Dimensions

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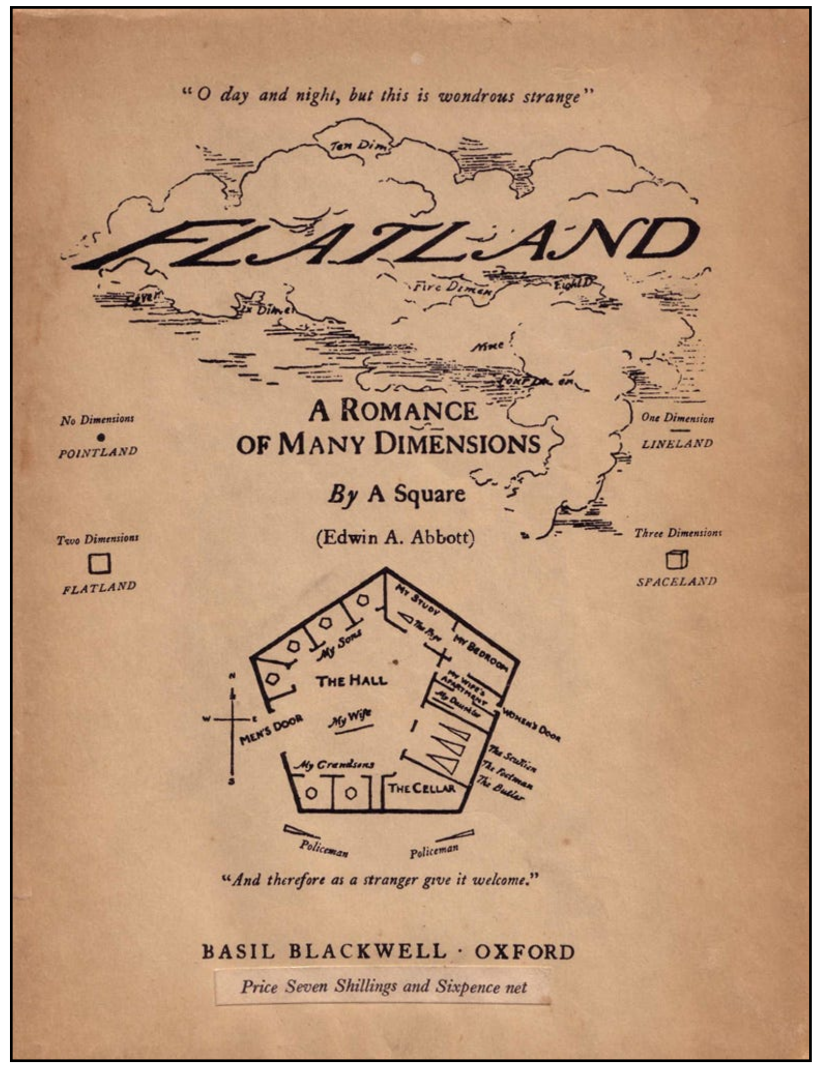

Flatland by Edwin A. Abbott

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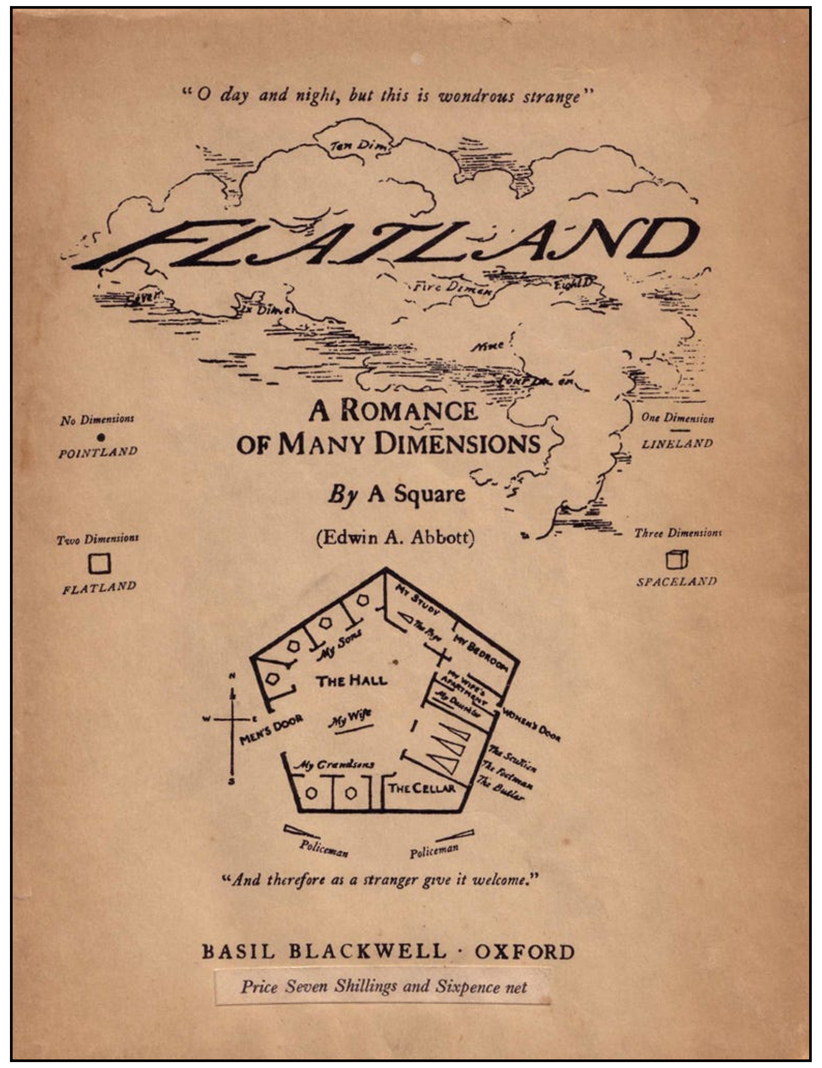

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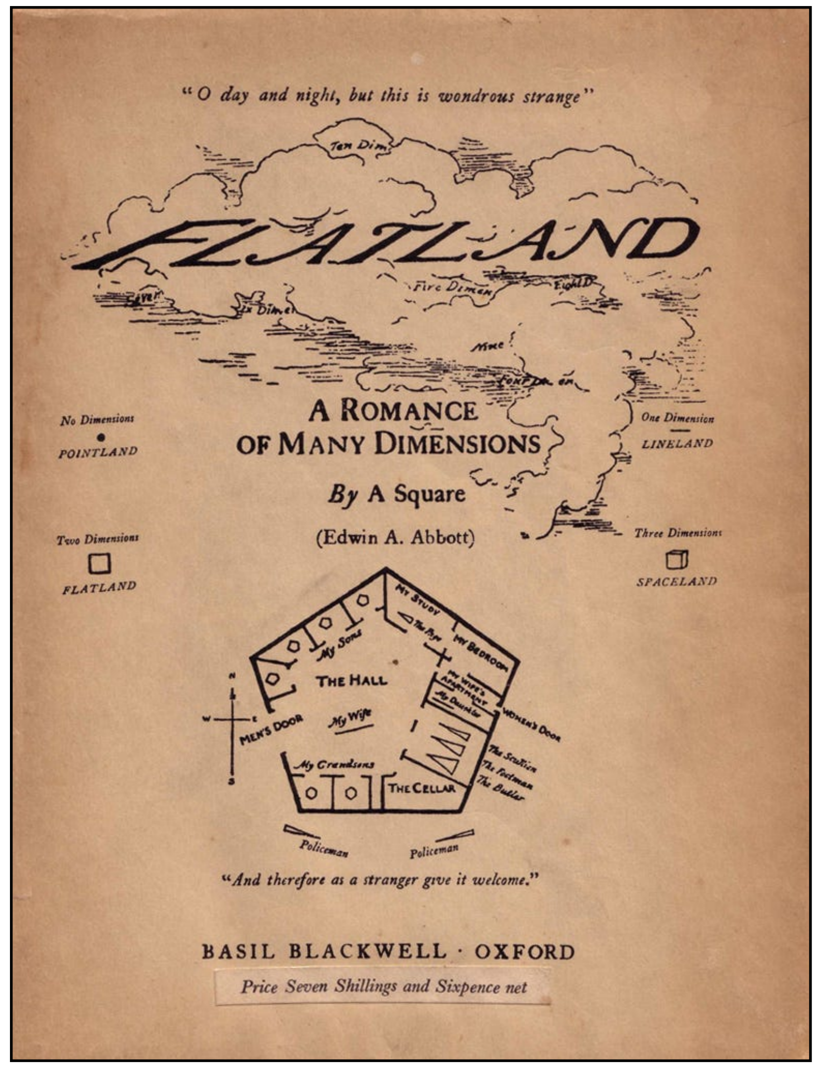
The moral. We have to be careful about our intuitions about higher-dimensional subspaces



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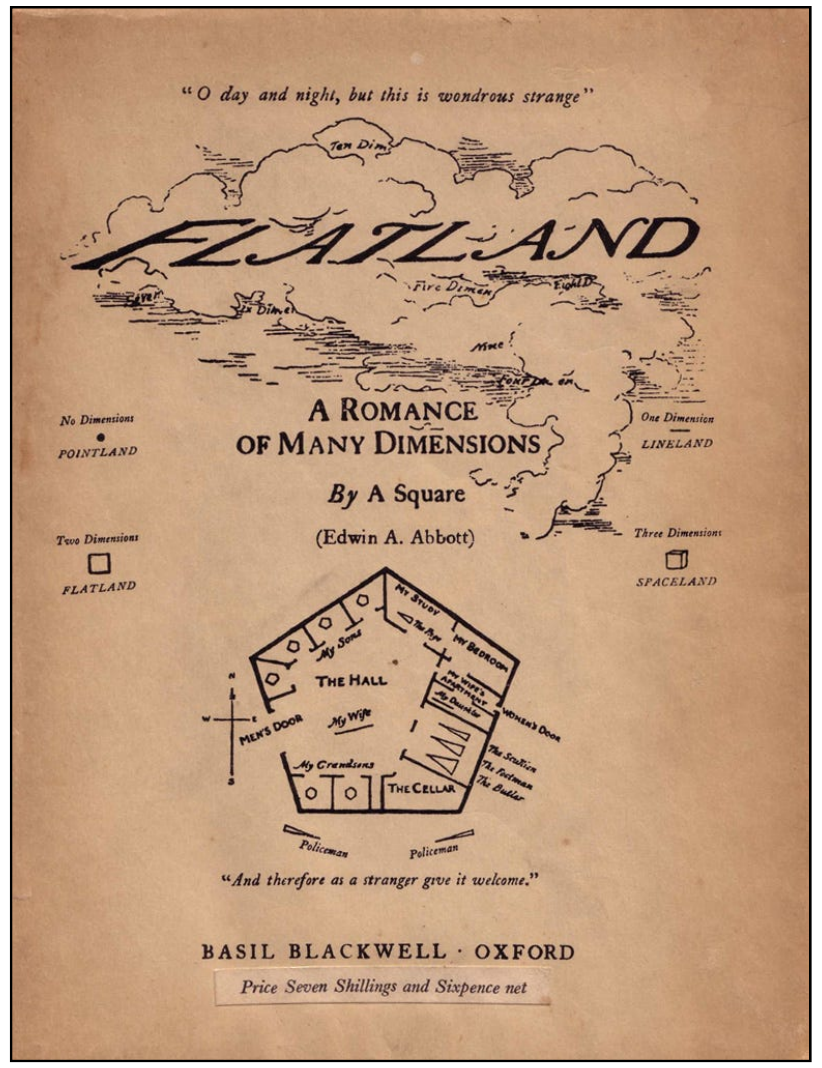
A 3D subspace of $\mathbb{R}^7$ "looks like" 3D space from the inside, but from the outside it may be "tilted"



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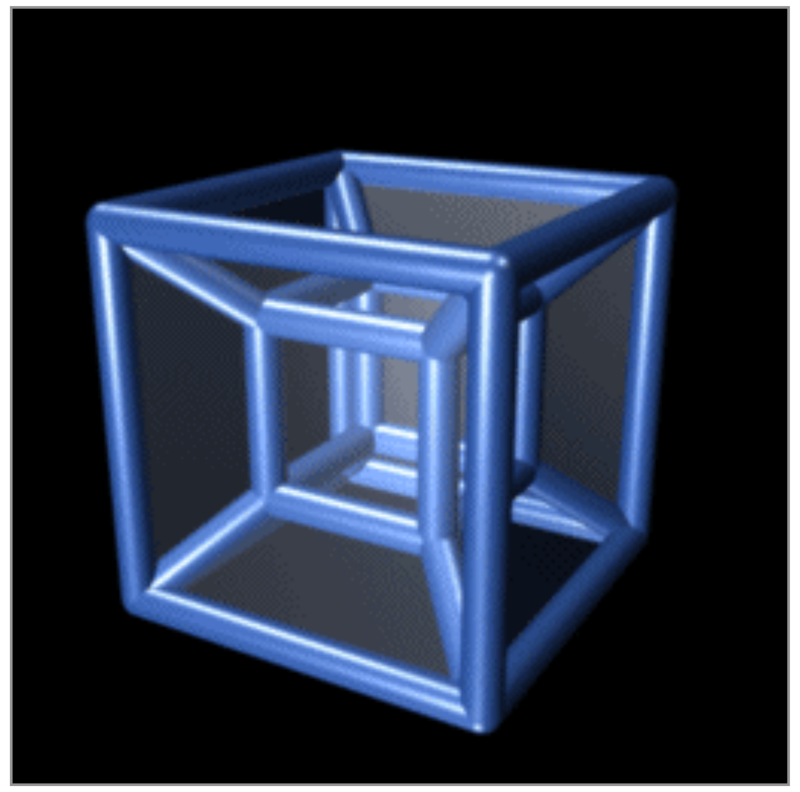

Projection of the 4D cube

Subspace (Algebraic Definition)

Definition. A **subspace** of $\mathbb{R}^n$ is a set H of vectors in $\mathbb{R}^n$ such that

1. for every $\mathbf{u}$ and $\mathbf{v}$ in H, the vector $\mathbf{u}+\mathbf{v}$ is in H

2. for every $\mathbf{u}$ in H and scalar c, the vector $c\mathbf{u}$ is in H

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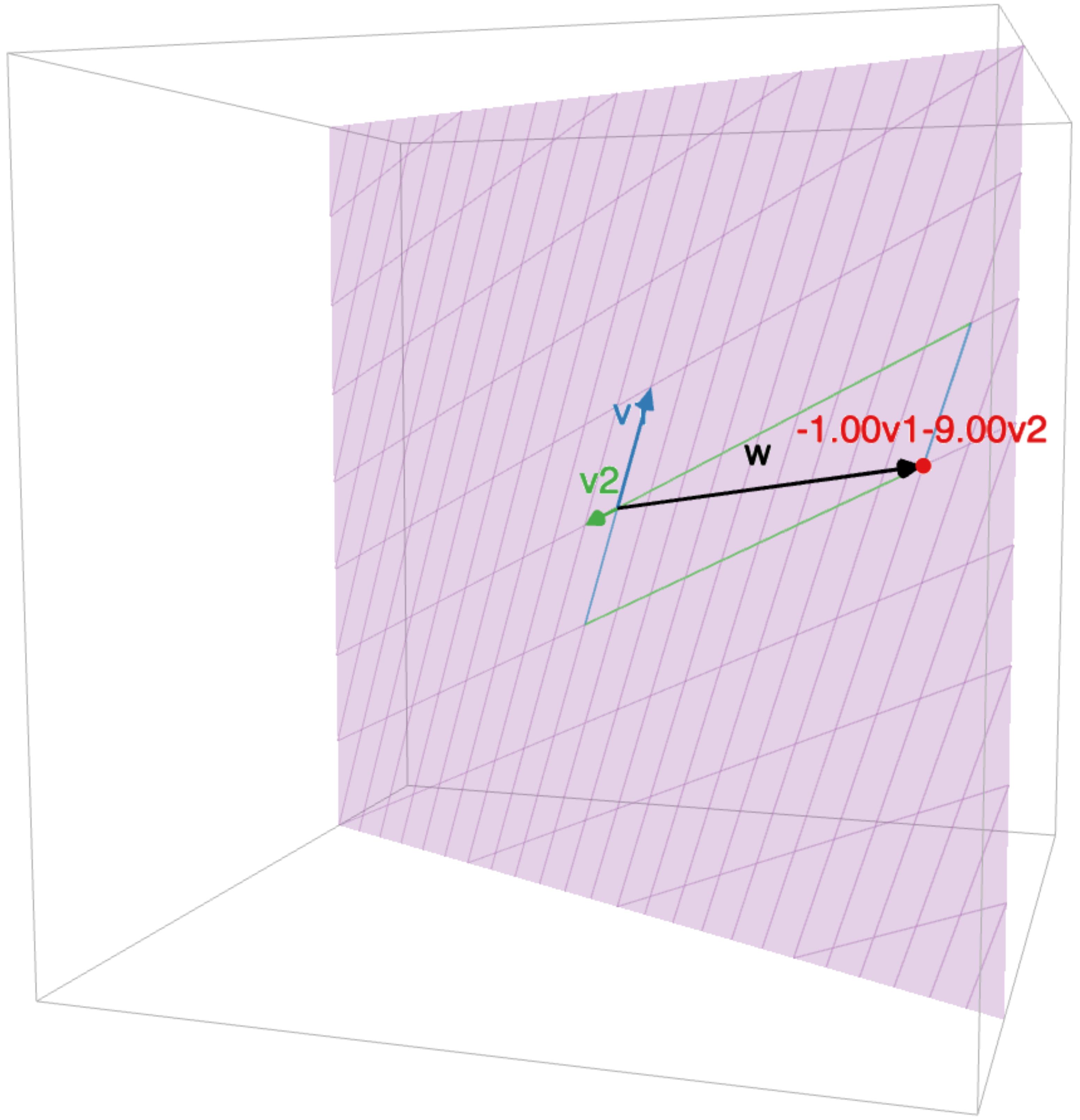
!! Subspaces must "live" somewhere !!

How to Think About this

It's not possible to "leave" H by addition or scaling

(recall this is also how we discussed spans)



https://textbooks.math.gatech.edu/ila/spans.html

How To: Verifying Subspaces

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Q. Verify that H is a subspace of $\mathbb{R}^n$

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1. Show if $\mathbf{u}$ and $\mathbf{v}$ are in H then so is $\mathbf{u}+\mathbf{v}$

2. Show if $\mathbf{u}$ is in H then so is $c\mathbf{u}$ for any scalar c

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How To: Verifying Non-Subspaces

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A.

Find $\mathbf{u}$ and $\mathbf{v}$ in H where $\mathbf{u}+\mathbf{v}$ is not in H

or

Find $\mathbf{u}$ in H where $c\mathbf{u}$ is not in H for *some* scalar c

Lines in $\mathbb{R}^2$

Subspaces must include the origin

Fact. For any subspace H of $\mathbb{R}^n$, the zero vector is in H. In set notation: $\mathbf{0} \in H$

(this can also be used to verify non-subspace-ness)

Example: The Origin

Fact. The set $\{\mathbf{0}_n\}$ is a subspace of $\mathbb{R}^n$

Example: $\mathbb{R}^n$

Fact. The set $\mathbb{R}^n$ is a subspace of $\mathbb{R}^n$

Example: Spans

Fact. For any set of vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ in $\mathbb{R}^n$ the set $\mathsf{span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a subspace of $\mathbb{R}^n$

Subspace in $\mathbb{R}^3$ (Geometrically)

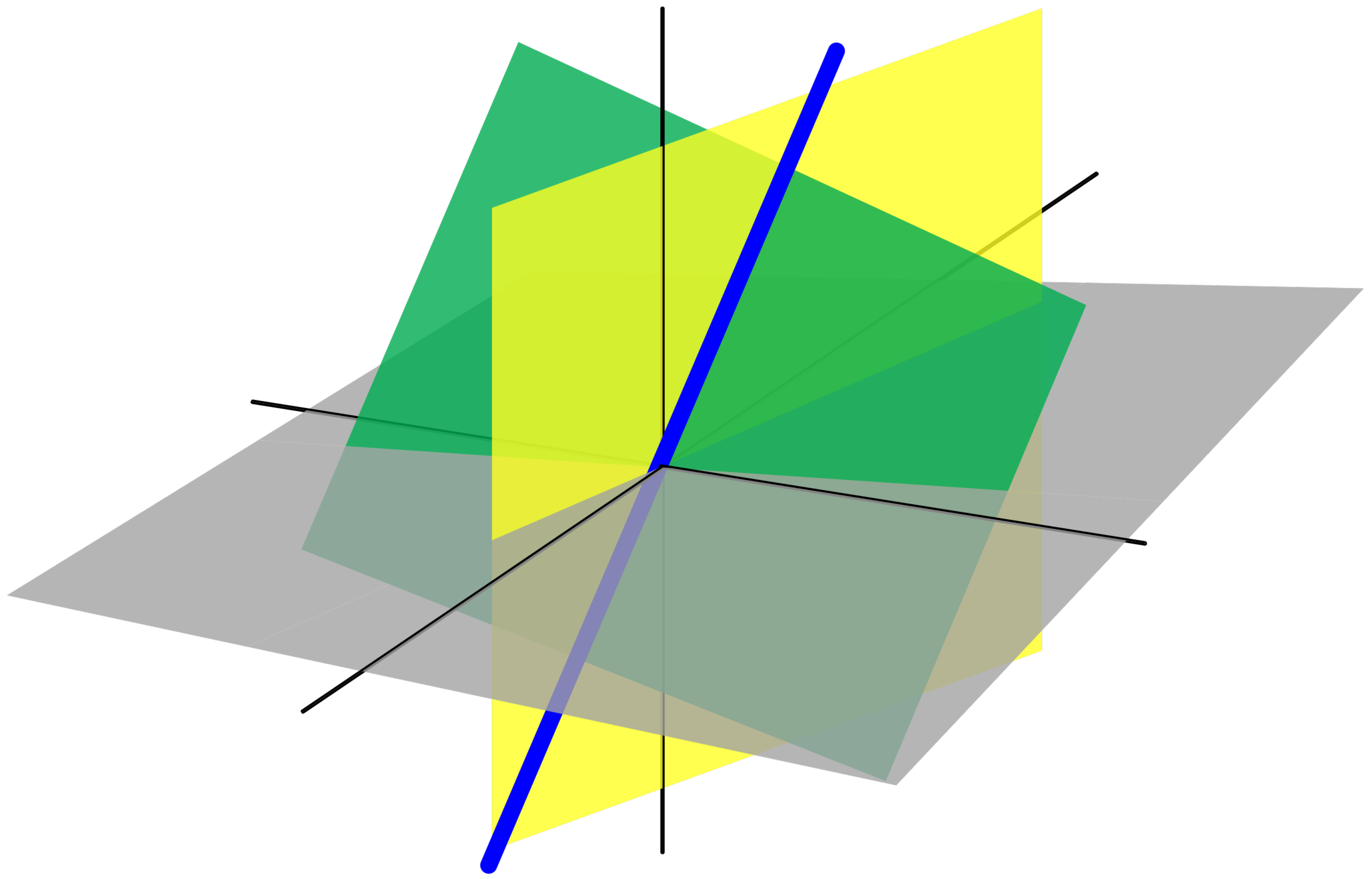
There are only 4 kinds of subspaces of $\mathbb{R}^3$:

1. $\{\mathbf{0}\}$ just the origin

2. lines (through the origin)

3. planes (through the origin)

4. All of $\mathbb{R}^3$



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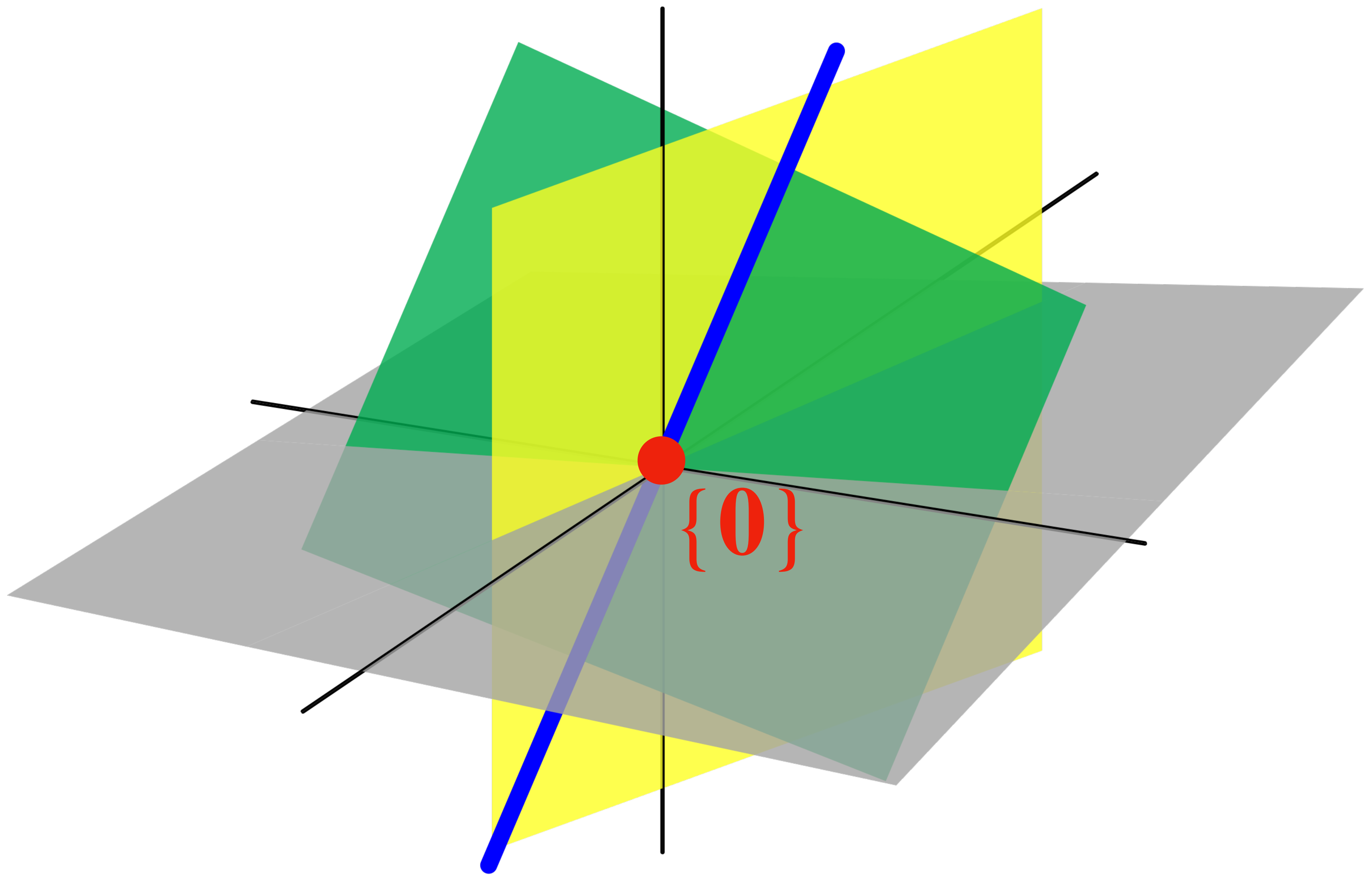

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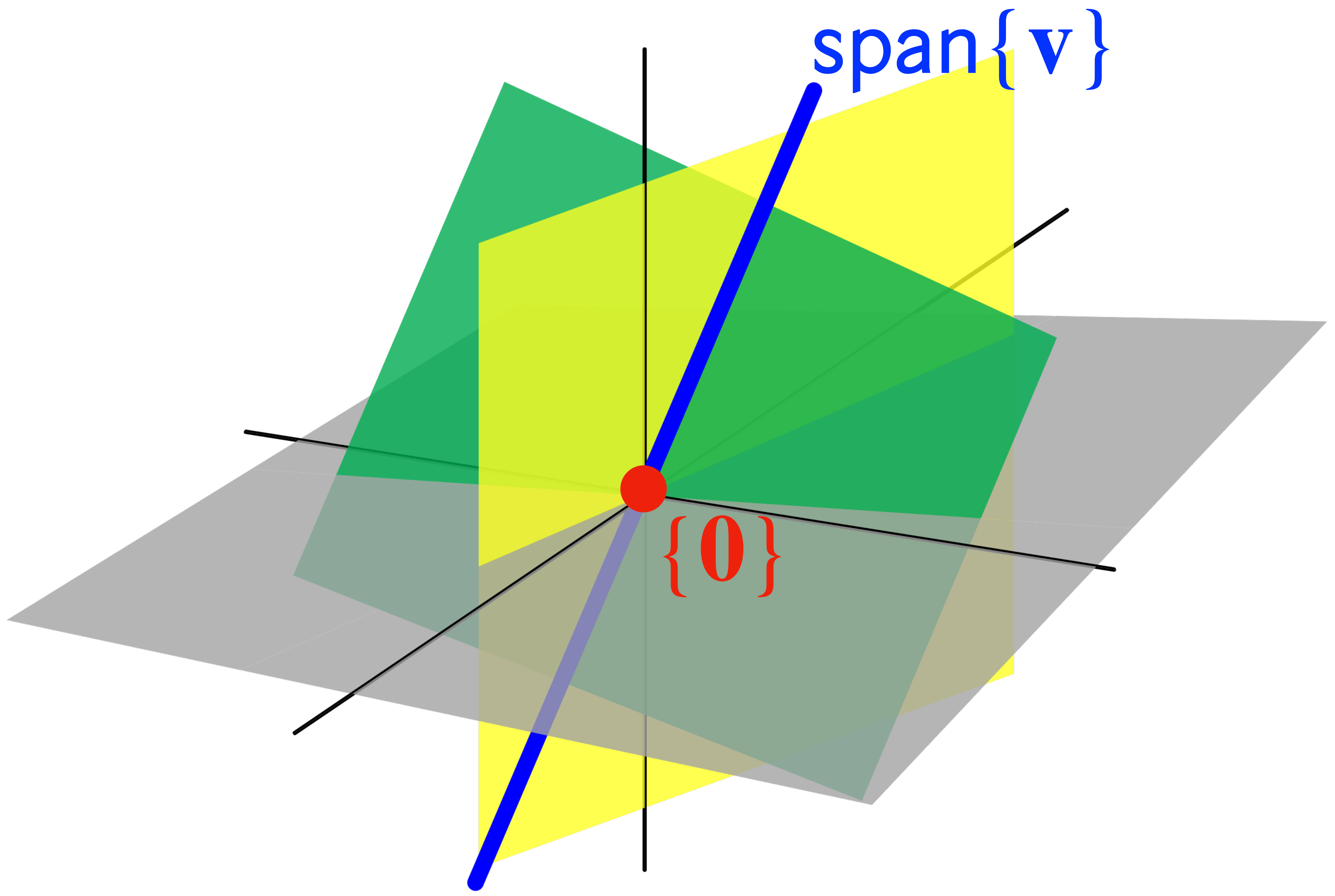

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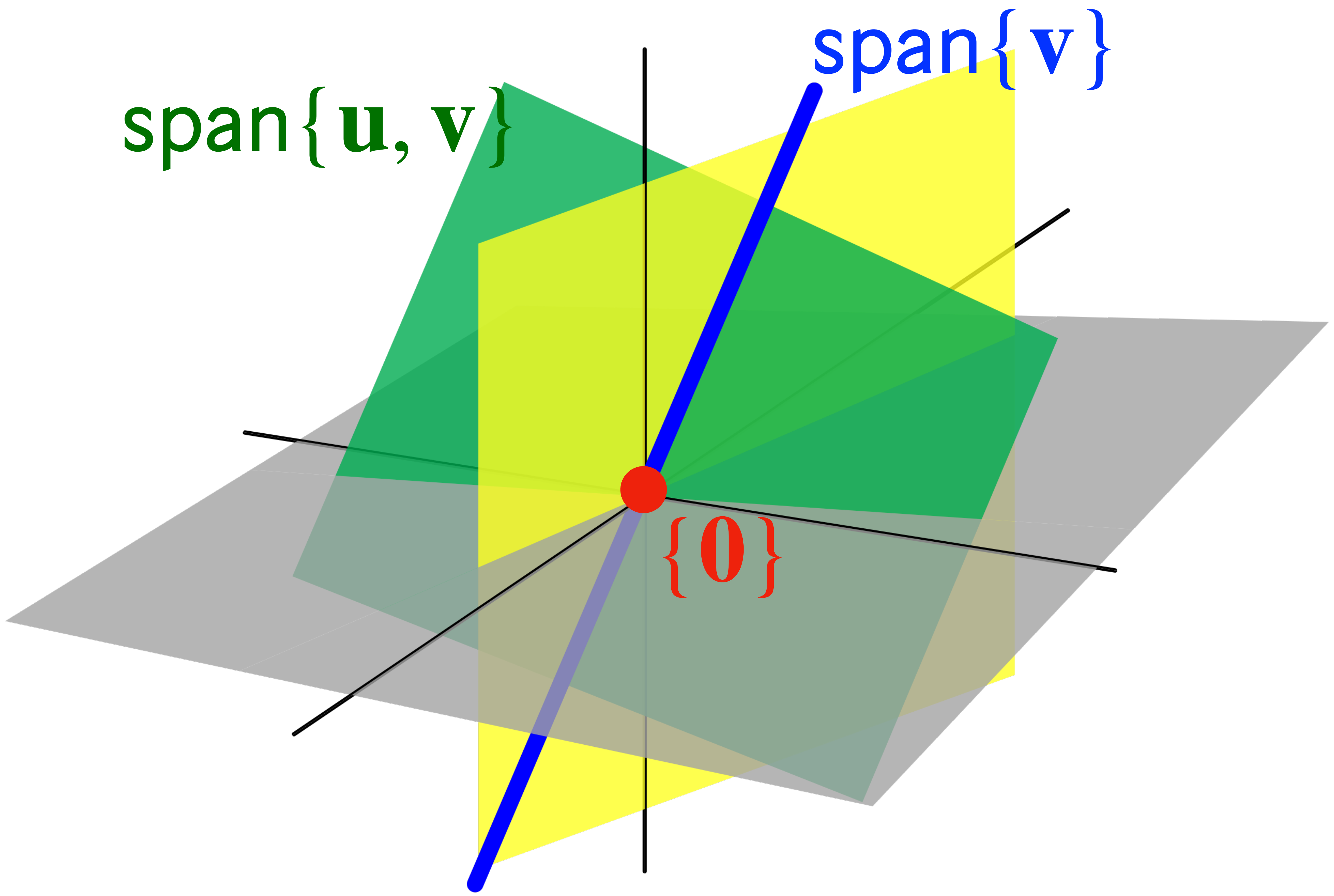

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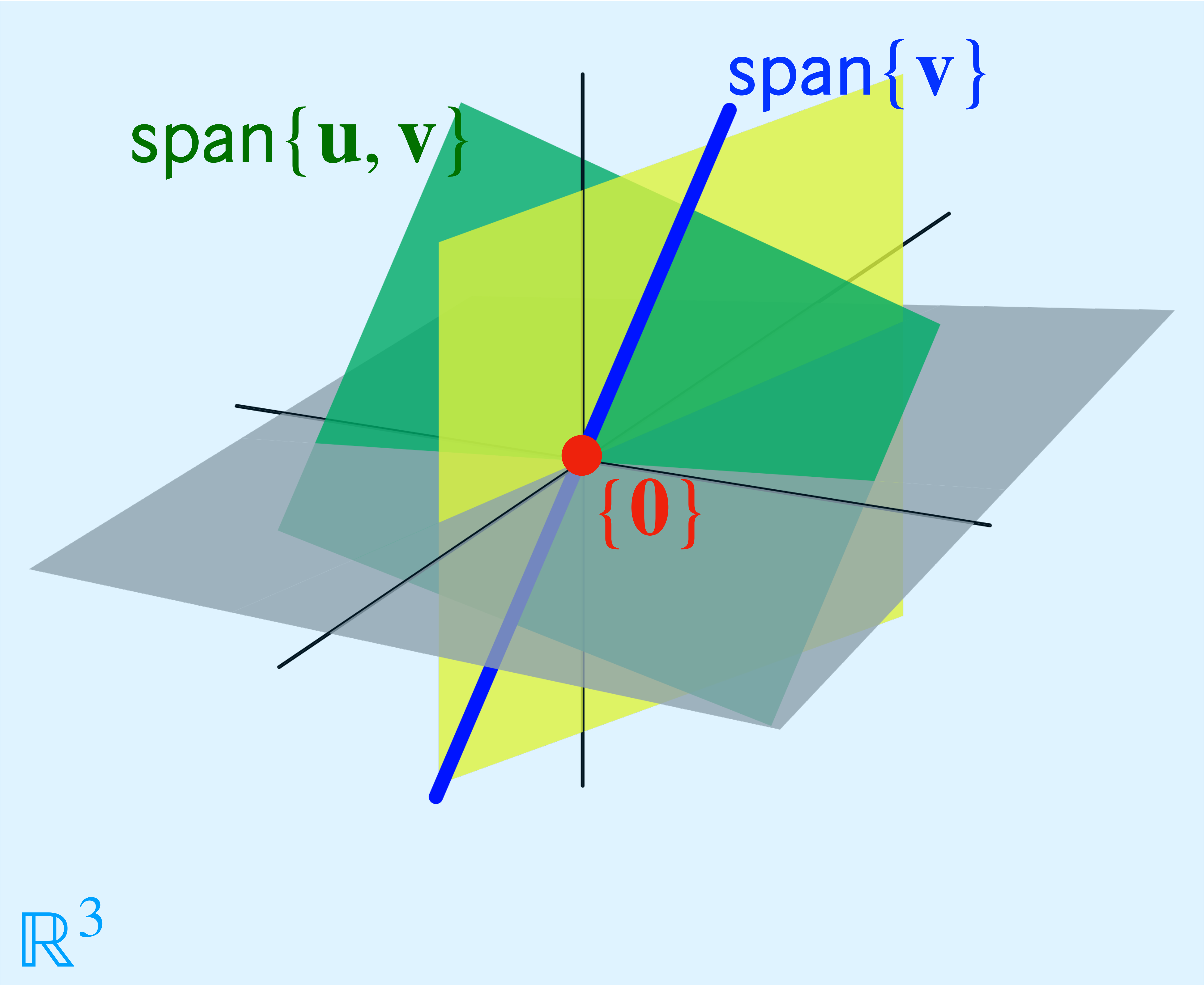

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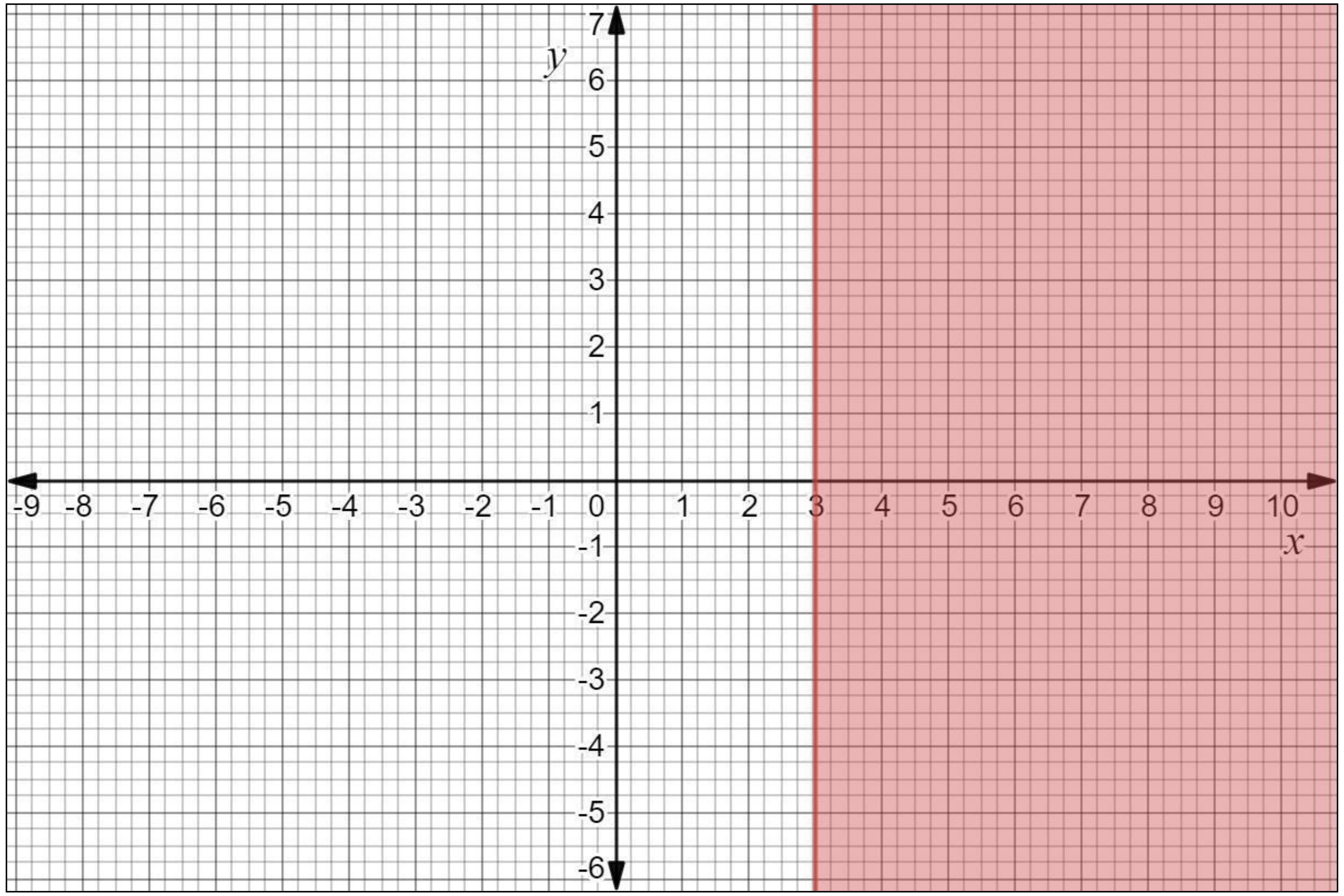
Non-Example: Bounded Sets

Fact. The set $\{(x, y) : x \geq 3\}$ is *not* a subspace of $\mathbb{R}^2$



https://brainly.com/question/14147114

Question

1. Show that the unit sphere $\{(x, y, z) : x^2 + y^2 + x^2 = 1\}$ *is* *not* *a subspace of* $\mathbb{R}^3$

2. Show that the range of a linear transformation $T : \mathbb{R}^m \to \mathbb{R}^n$ *is a subspace of* $\mathbb{R}^n$

Answer (1)

Answer (2)

How To: Subspaces and Span

Q. Show that $\mathbf{v}$ lies in the subspace generated by $\mathbf{u}_1, \ldots, \mathbf{u}_k$

A. Show that $\mathbf{v}$ is in $\text{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$

We will start using "subspace generated by" and "span of" interchangeably

Subspaces and Matrices

The Connection

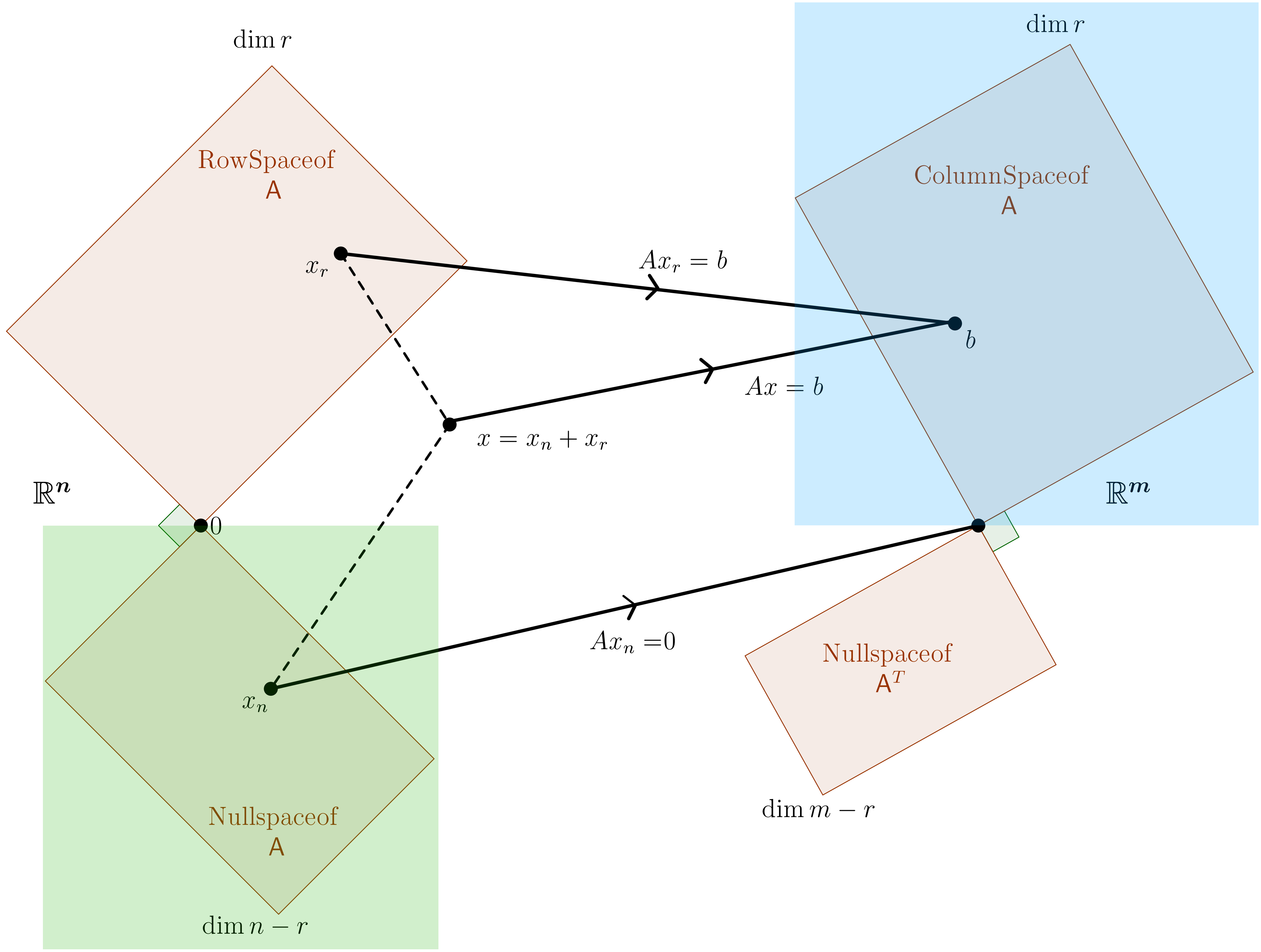
Since matrices can be viewed as:

- collections of vectors
- implementing linear transformations

they have many associated subspaces

Today we'll look at:

- column space
- null space



https://mathworld.wolfram.com/FundamentalMatrixSubspaces.html

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Df. *The column space of a matrix is the span of its columns*

Df. *The column space of a matrix is the* <u>*range*</u> *of the linear transformation it implements*

Subspace of What?

$$\begin{array}{c|c} & n \\ \hline m & \begin{bmatrix} | & | & \cdots & | & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_{n-1} & \mathbf{a}_n \\ | & | & \cdots & | & | \end{bmatrix} \end{array}$$

$c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + \ldots c_n\mathbf{a}_n$ **is a vector in** $\mathbb{R}^m$

$\text{Col}(A)$

is a subspace of

$\mathbb{R}^m$

Examples

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & -1 & 4 \\ 2 & -2 & 5 \\ 3 & -3 & 6 \end{bmatrix}$$

Null Space

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The null space of a matrix A is the set of all vectors that are mapped to the zero vector by A

Subspace of What?

$$\begin{array}{r|c} & \overline{\quad n \text{ columns} \quad} \\ m \text{ rows} & A\mathbf{v} = \mathbf{0} \end{array}$$

$m \times n \quad n \times 1 \quad\quad m \times 1$

$\mathbf{v}$ is a vector in $\mathbb{R}^n$

$\text{Nul}(A)$

is a subspace of

$\mathbb{R}^n$

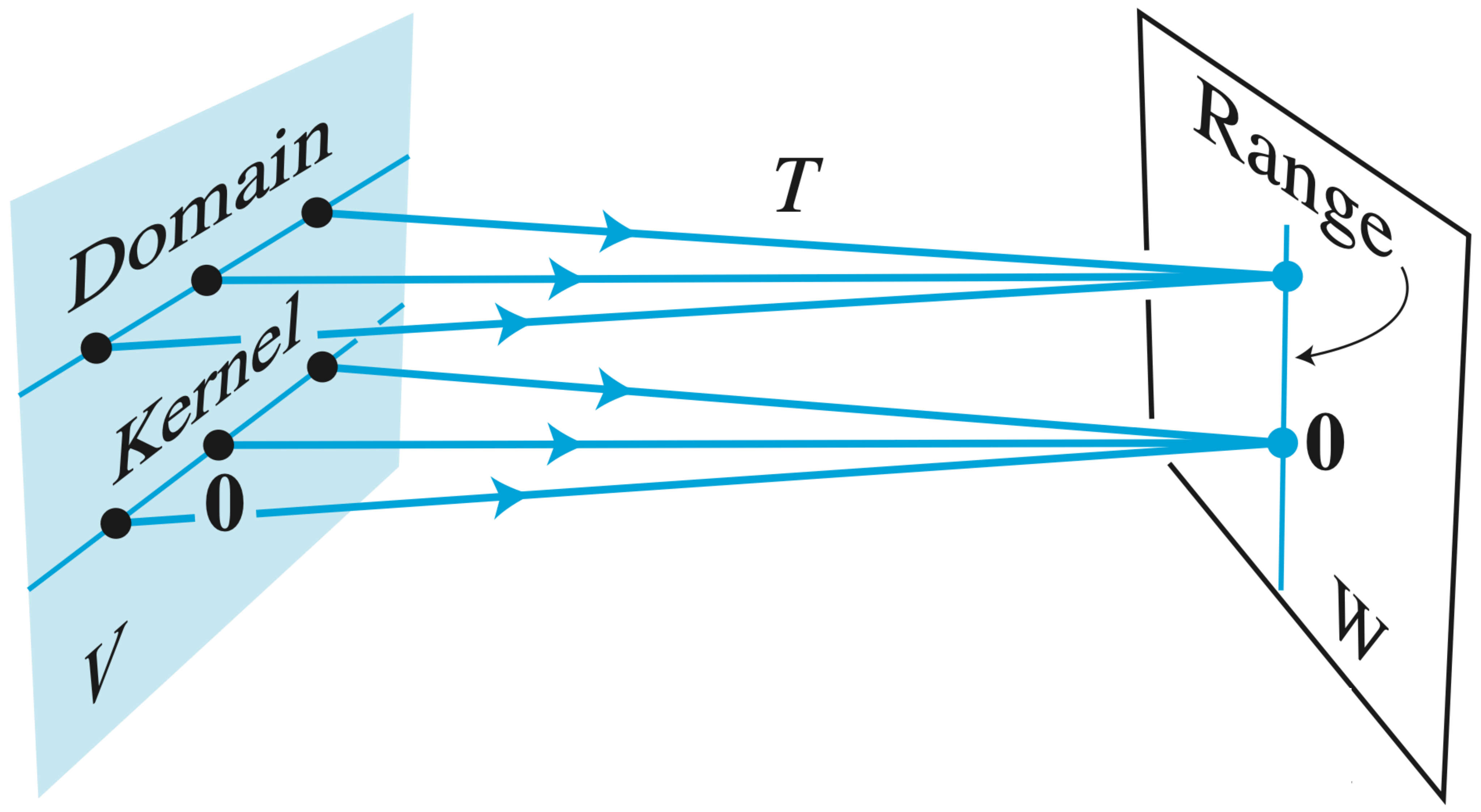
The Null Space is a Subspace

Fact. For any $m \times n$ matrix A, the set $\mathrm{Nul}(A)$ is a subspace of $\mathbb{R}^n$

Examples

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 0 & 0 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & -1 & 4 \\ 2 & -2 & 5 \\ 3 & -3 & 6 \end{bmatrix}$$

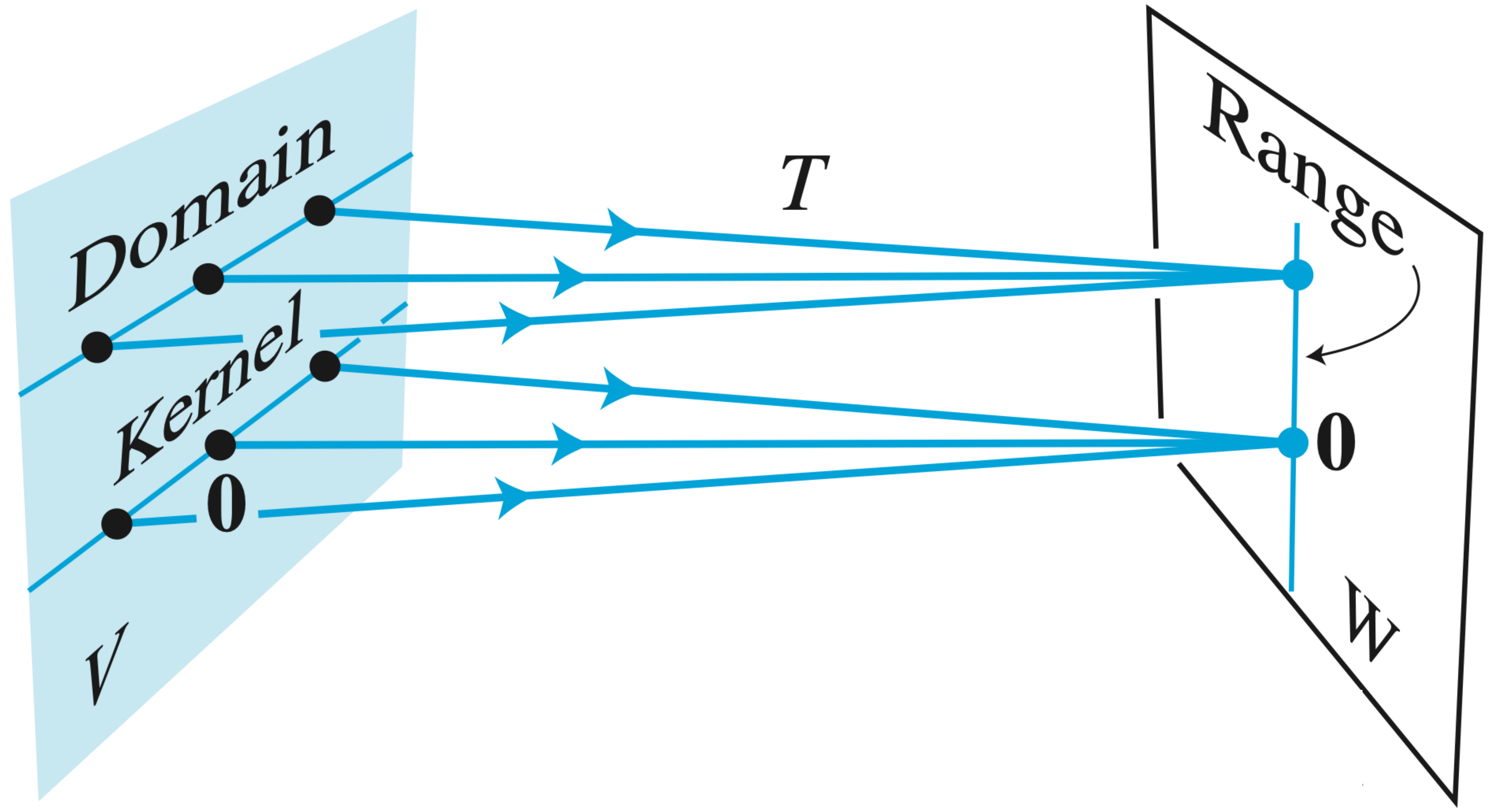
Linear Transformations Perspective



Linear Algebra and its Applications (Lay, Lay, McDonald)

Linear Transformations Perspective

If A implements T then



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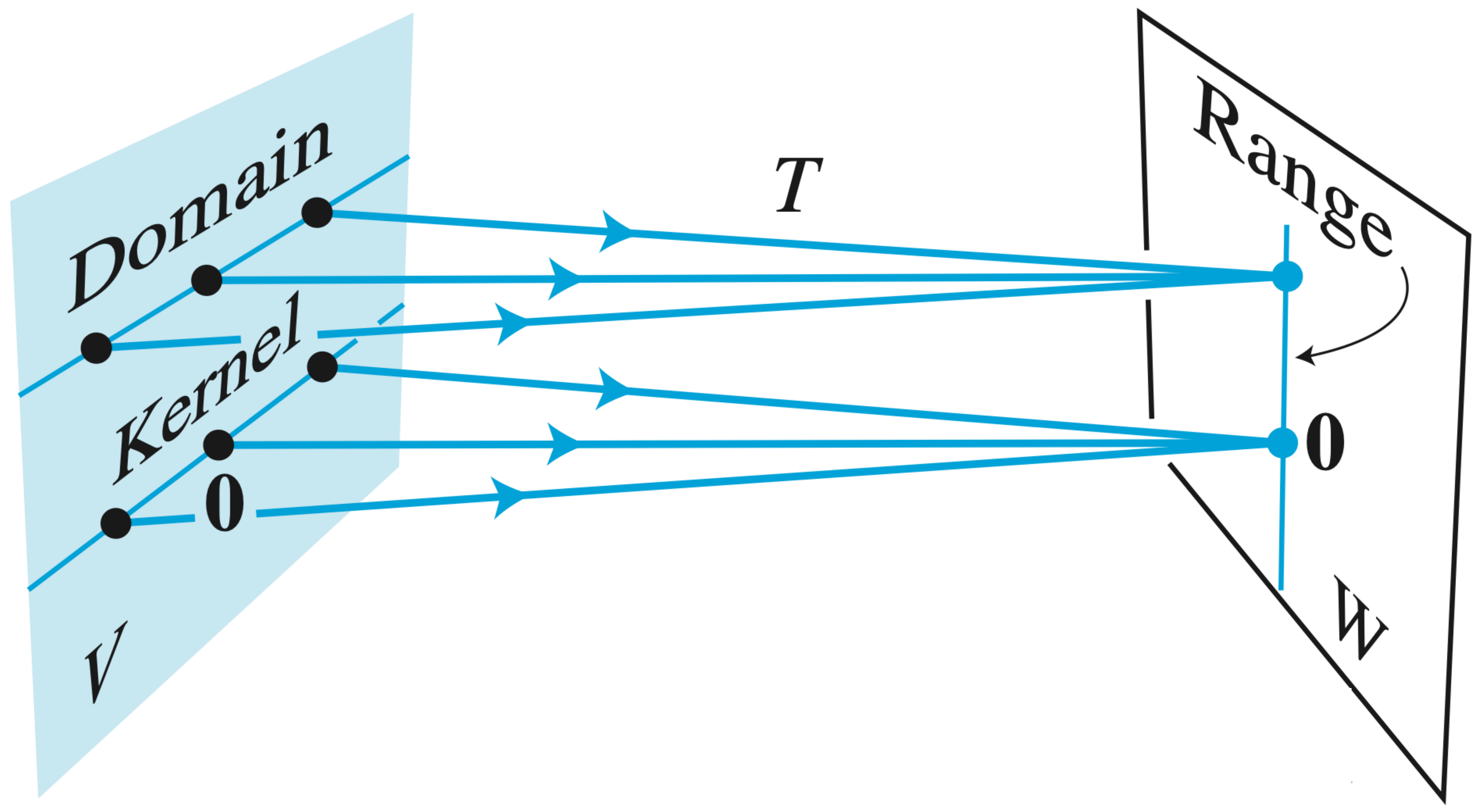
If A implements T then

» $\mathrm{Col}(A)$ is the same as $\mathrm{ran}(T)$, where vectors are "sent" by T

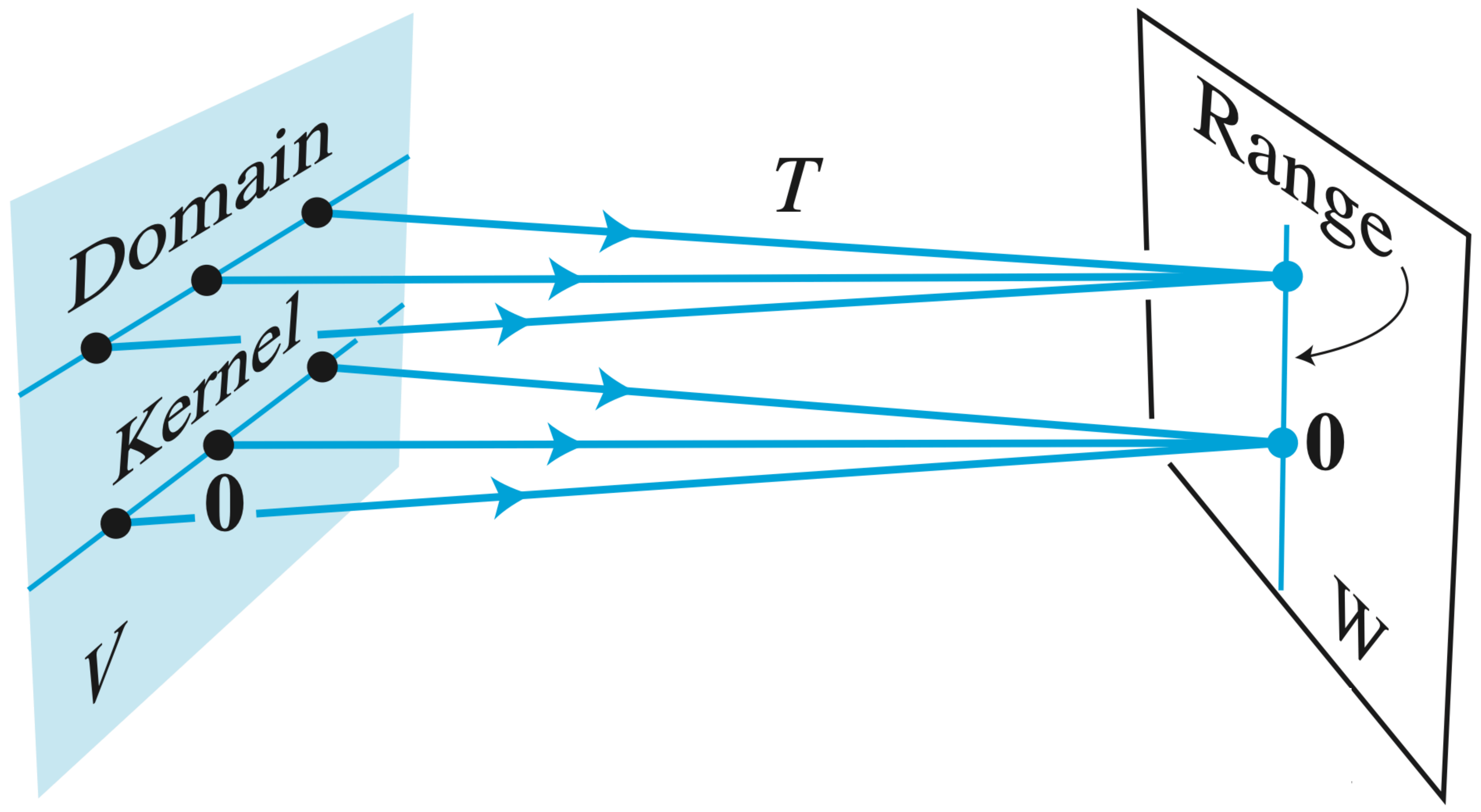


Linear Transformations Perspective

If A implements T then

» $\mathrm{Col}(A)$ is the same as $\mathrm{ran}(T)$, where vectors are "sent" by T

» $\mathrm{Nul}(A)$ is the set of vectors that are "zeroed out" by T, also called the **kernel** of T



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Comparing Column Space and Null Space

The column space and the null space live can live in entirely different spaces

The point. They are not easily comparable

Contrast Between Nul A and Col A for an m x n Matrix A

Nul A	Col A
1. Nul A is a subspace of $\mathbb{R}^n$.	**1.** Col A is a subspace of $\mathbb{R}^m$.
2. Nul A is implicitly defined; that is, you are given only a condition ($A\mathbf{x} = \mathbf{0}$) that vectors in Nul A must satisfy.	**2.** Col A is explicitly defined; that is, you are told how to build vectors in Col A.
3. It takes time to find vectors in Nul A. Row operations on $[\,A \quad \mathbf{0}\,]$ are required.	**3.** It is easy to find vectors in Col A. The columns of A are displayed; others are formed from them.
4. There is no obvious relation between Nul A and the entries in A.	**4.** There is an obvious relation between Col A and the entries in A, since each column of A is in Col A.
5. A typical vector $\mathbf{v}$ in Nul A has the property that $A\mathbf{v} = \mathbf{0}$.	**5.** A typical vector $\mathbf{v}$ in Col A has the property that the equation $A\mathbf{x} = \mathbf{v}$ is consistent.
6. Given a specific vector $\mathbf{v}$, it is easy to tell if $\mathbf{v}$ is in Nul A. Just compute $A\mathbf{v}$.	**6.** Given a specific vector $\mathbf{v}$, it may take time to tell if $\mathbf{v}$ is in Col A. Row operations on $[\,A \quad \mathbf{v}\,]$ are required.
7. Nul $A = \{\mathbf{0}\}$ if and only if the equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.	**7.** Col $A = \mathbb{R}^m$ if and only if the equation $A\mathbf{x} = \mathbf{b}$ has a solution for every $\mathbf{b}$ in $\mathbb{R}^m$.
8. Nul $A = \{\mathbf{0}\}$ if and only if the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.	**8.** Col $A = \mathbb{R}^m$ if and only if the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ *onto* $\mathbb{R}^m$.

(just for reference)

Linear Algebra and its Applications (Lay, Lay, McDonald)

Bases

The idea behind a basis

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Every subspace is the span of a set of vectors

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Every subspace is the span of a set of vectors

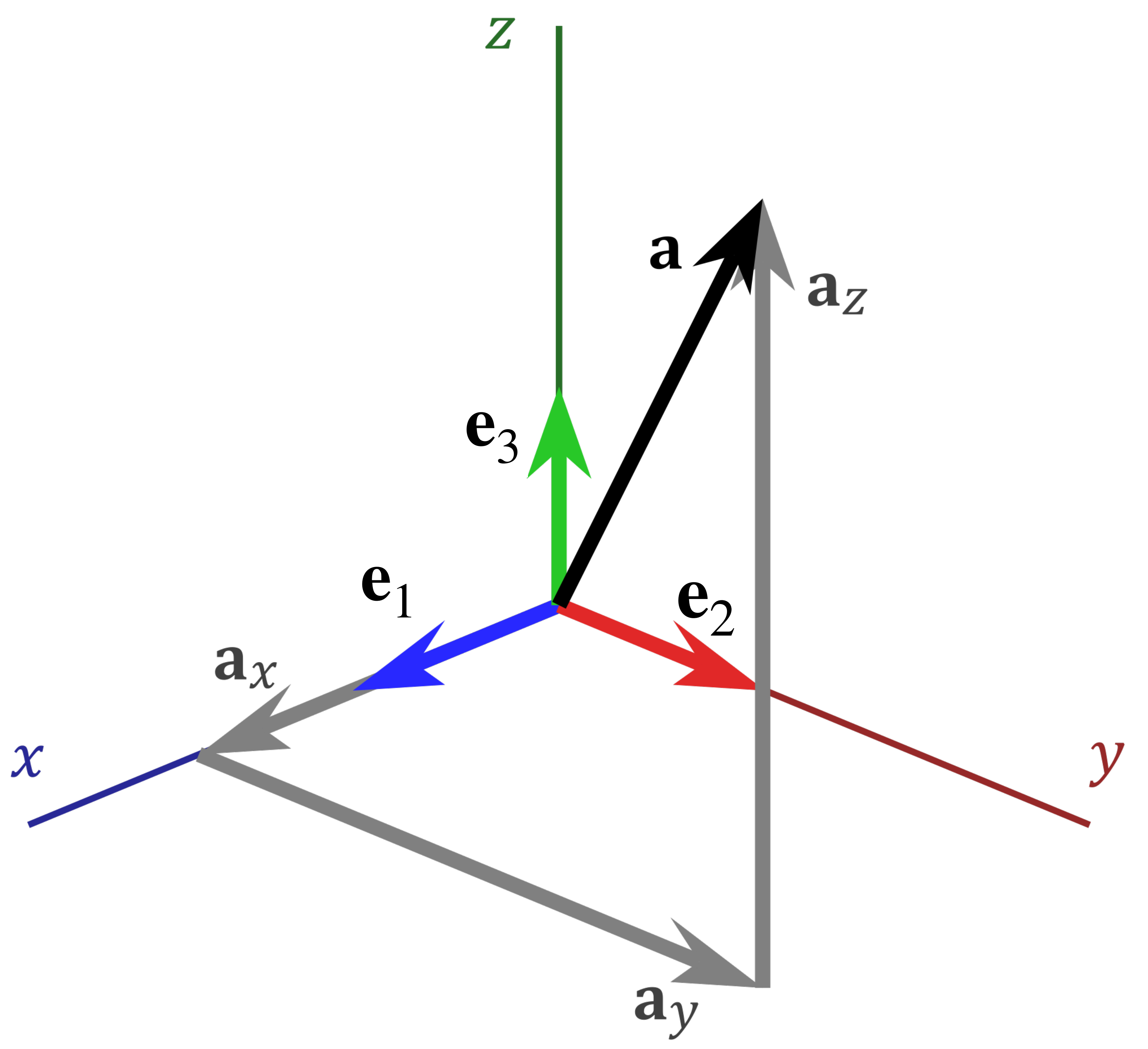
A basis is a "minimal" choice of these vectors, a "concise representation" of a subspace

Recall: Standard Basis

Df. The ***n–dimensional standard basis vectors*** (or standard coordinate vectors) are the vectors $\mathbf{e}_1, \ldots, \mathbf{e}_n$ where

$$\mathbf{e}_i = \begin{array}{cc} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix} & \begin{matrix} 1 \\ 2 \\ \vdots \\ i-1 \\ i \\ i+1 \\ \vdots \\ n-1 \\ n \end{matrix} \end{array}$$

What's special about the standard basis?



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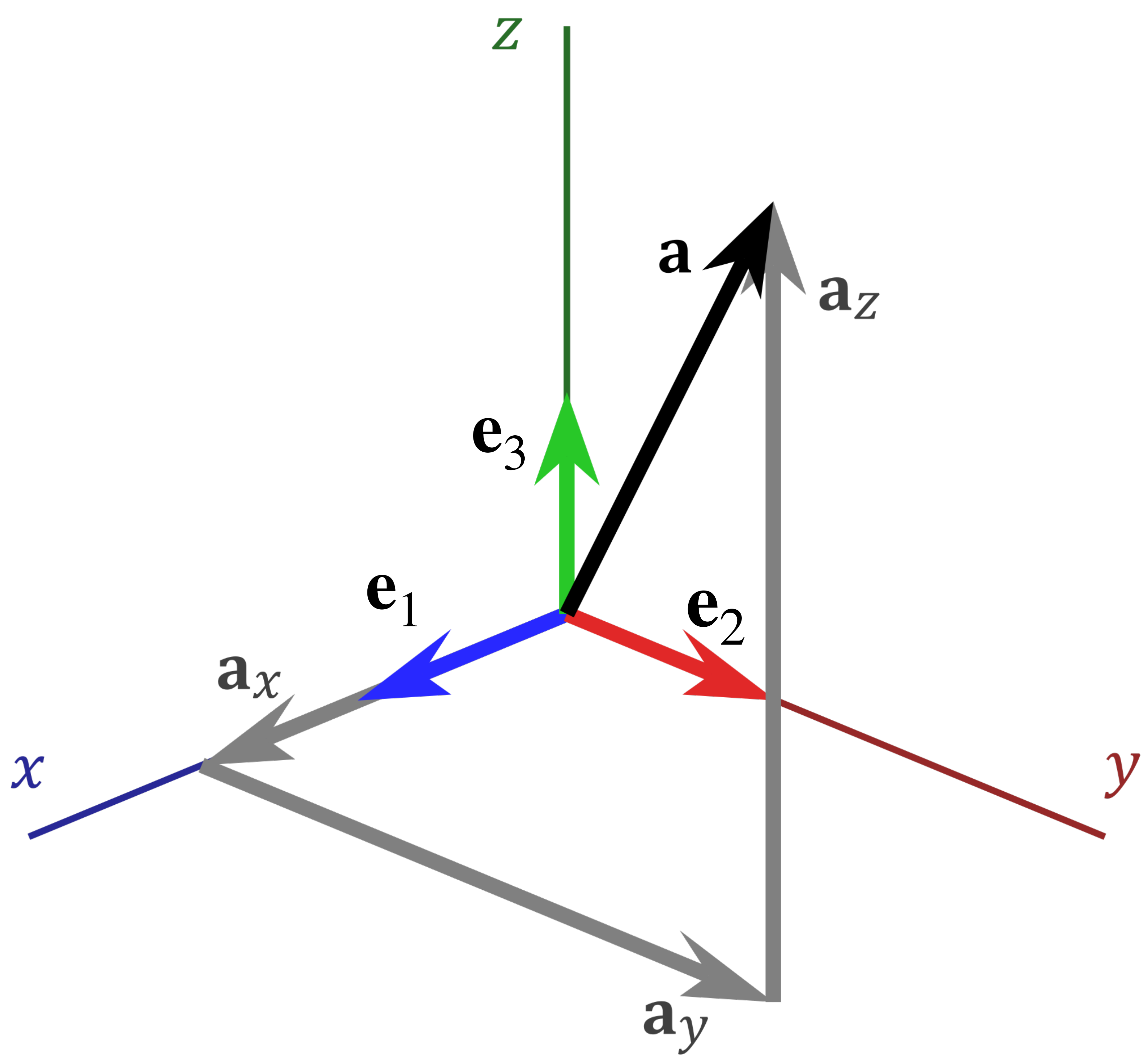
The n standard basis vectors in $\mathbb{R}^n$:

» are linearly independent
» span all of $\mathbb{R}^n$

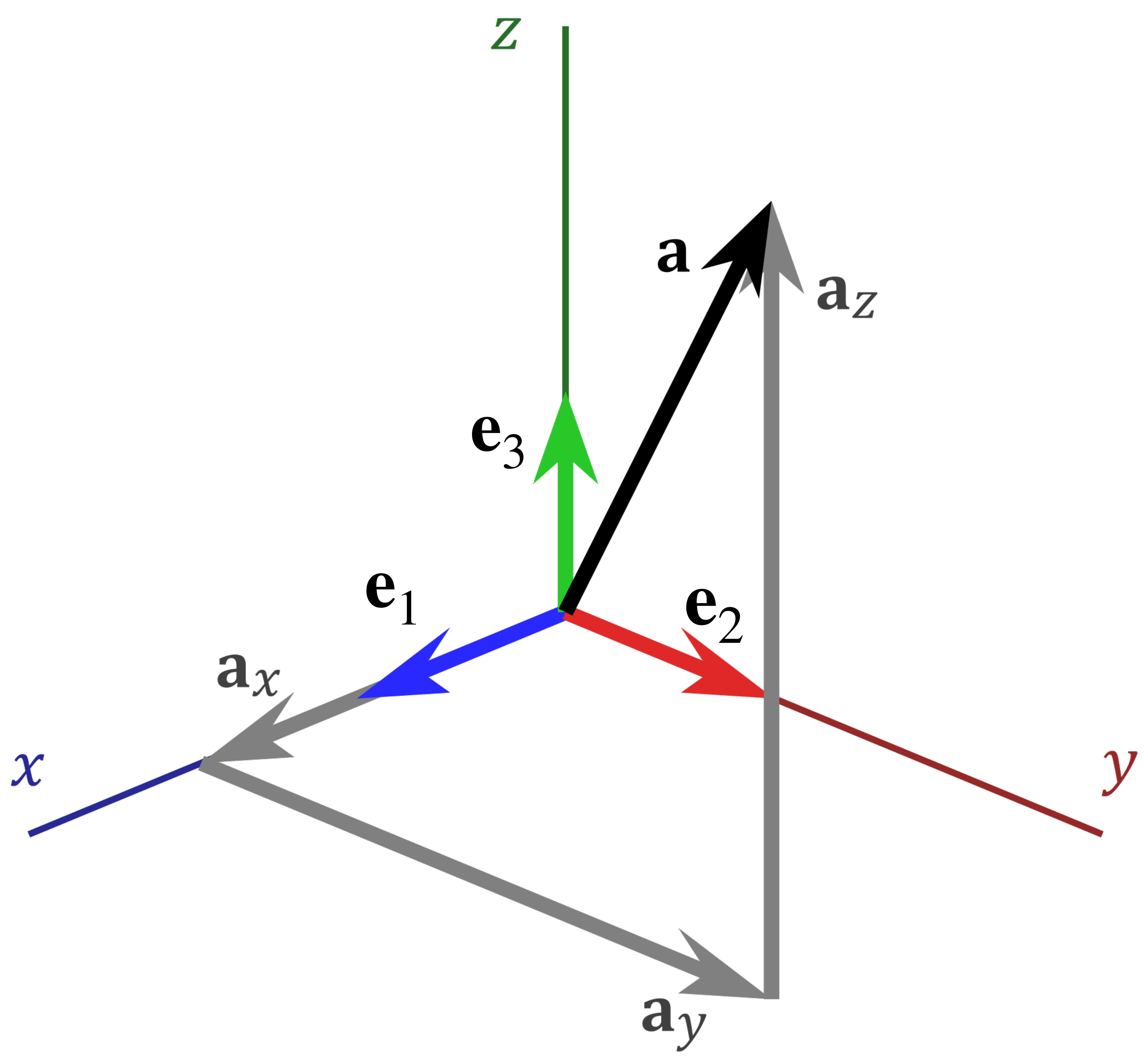


What's special about the standard basis?

The n standard basis vectors in $\mathbb{R}^n$:

» are linearly independent
» span all of $\mathbb{R}^n$

Their span is as "large" as possible while the set of vectors generating the span is as "small" as possible



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A basis is a *minimal* set of vectors which spans all of H

Example: Standard basis

The standard basis is a basis of $\mathbb{R}^n$

$$\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + \ldots + v_n\mathbf{e}_n$$

Column vectors are just weights for a linear combination of the standard basis

Example: Column Space of Invertible Matrices

Fact. The columns of an invertible $n \times n$ matrix form a basis of $\mathbb{R}^n$

Subsets of Spanning Sets

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Thm. If the vectors $\mathbf{v}_1$, $\mathbf{v}_2$,...,$\mathbf{v}_k$ span a subspace H then a subset of them form a basis of H

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We can *remove* vectors from a spanning set until we get a basis

How do we do this?

As usual, by connecting back to matrices

Question

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ -3 \end{bmatrix}, \begin{bmatrix} -1 \\ -2 \\ 3 \end{bmatrix} \right\}$$

Is this set of vectors a basis for $\mathbb{R}^3$*?*

Bases of Column Space and Null Space

The Goal of this Last Section

Determine how to find bases for the **column space** and the **null space** of a given matrix

How to: Finding a basis for the null space

Q. Given a $m \times n$ matrix A find a basis for $\mathrm{Nul}(A)$

How to: Finding a basis for the null space

Q. Given a $m \times n$ matrix A find a basis for $\mathrm{Nul}(A)$

The idea. Describe the solutions of $A\mathbf{x} = 0$ as linear combination of vectors

Example

$$A \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Let's write down a general form solution for A:

Parametric Solutions

We can think of our general form solution as a *linear transformation*

$x_1 = 2x_2 + x_4 - 3x_5$

x_2 is free

$x_3 = (-2)x_4 + 2x_5$

x_4 is free

x_5 is free

$\equiv$

"given values for x_2, x_3, and x_4, I can give you a solution"

Parametric Solutions

We can think of our general form solution as a *linear transformation*

$x_1 = 2x_2 + x_4 - 3x_5$

$\boxed{x_2}$ is free

$x_3 = (-2)x_4 + 2x_5$

$\boxed{x_4}$ is free

$\boxed{x_5}$ is free

$$\equiv$$

$$\begin{bmatrix} s \\ t \\ u \end{bmatrix} \mapsto \begin{bmatrix} 2s + t - 3u \\ s \\ (-2)t + 2u \\ t \\ u \end{bmatrix}$$

Parametric Solutions

We can think of our general form solution as a *linear transformation*

!! this transformation is only linear !!
!! in the case of homogeneous equations !!

$x_1 = 2x_2 + x_4 - 3x_5$
x_2 is free
$x_3 = (-2)x_4 + 2x_5$
x_4 is free
x_5 is free

$$\equiv \quad \begin{bmatrix} s \\ t \\ u \end{bmatrix} \mapsto \begin{bmatrix} 2s + t - 3u \\ s \\ (-2)t + 2u \\ t \\ u \end{bmatrix}$$

Example

$$\begin{bmatrix} s \\ t \\ u \end{bmatrix} \mapsto \begin{bmatrix} 2s + t - 3u \\ s \\ (-2)t + 2u \\ t \\ u \end{bmatrix}$$

Let's find the matrix implementing this transformation:

Example

$$\begin{bmatrix} 2 & 1 & -3 \\ 1 & 0 & 0 \\ 0 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Example

$$\begin{bmatrix} 2 & 1 & -3 \\ 1 & 0 & 0 \\ 0 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Every solution to $A\mathbf{x} = \mathbf{0}$ can be written as an *image* of this transformation

Example

$$\begin{bmatrix} 2 & 1 & -3 \\ 1 & 0 & 0 \\ 0 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Every solution to $A\mathbf{x} = \mathbf{0}$ can be written as an *image* of this transformation

So every solution can be written as a linear combination of its *columns*

Example

$$\begin{bmatrix} 2 & 1 & -3 \\ 1 & 0 & 0 \\ 0 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Every solution to $A\mathbf{x} = \mathbf{0}$ can be written as an *image* of this transformation

So every solution can be written as a linear combination of its *columns*

The columns of this matrix span $\mathrm{Nul}(A)$

Example

$$\begin{bmatrix} 2 & 1 & -3 \\ 1 & 0 & 0 \\ 0 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Thm. The columns of this matrix are linearly independent

Example

$$\begin{bmatrix} 2 & 1 & -3 \\ 1 & 0 & 0 \\ 0 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Example

$$\begin{bmatrix} 2 & 1 & -3 \\ 1 & 0 & 0 \\ 0 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The columns of this matrix *span* $\mathsf{Nul}(A)$

Example

$$\begin{bmatrix} 2 & 1 & -3 \\ 1 & 0 & 0 \\ 0 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The columns of this matrix *span* $\text{Nul}(A)$

The columns of this matrix are linearly independent

Example

$$\begin{bmatrix} 2 & 1 & -3 \\ 1 & 0 & 0 \\ 0 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The columns of this matrix *span* $\mathrm{Nul}(A)$

The columns of this matrix are linearly independent

The columns of this matrix form a basis for $\mathrm{Nul}(A)$

Example

Alternatively, we can write a general form solution so that it is a linear combination of vectors with *free variables as weights*

$x_1 = 2x_2 + x_4 - 3x_5$

x_2 is free

$x_3 = (-2)x_4 + 2x_5$

x_4 is free

x_5 is free

How to: Finding a basis for the null space

Q. Given a $m \times n$ matrix A find a basis for $\mathrm{Nul}(A)$

A.

1. Find a general form solution for $A\mathbf{x} = \mathbf{0}$
2. Write this solution as a linear combination of vectors *where the free variables are the weights*
3. The resulting vectors form a basis for $\mathrm{Nul}(A)$

An Observation

The *number* of vectors in the basis we found is the same as the number of *free variables* in a general form solution

$x_1 = 2x_2 + x_4 - 3x_5$
x_2 is free
$x_3 = (-2)x_4 + 2x_5$
x_4 is free
x_5 is free

$$\equiv \quad \begin{bmatrix} s \\ t \\ u \end{bmatrix} \mapsto \begin{bmatrix} 2s + t - 3u \\ s \\ (-2)t + 2u \\ t \\ u \end{bmatrix}$$

Practice Problem

$$A \sim \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

Determine a basis for $\mathrm{Nul}(A)$

Answer

$$\begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

onto column space...

How To: Finding a basis for the column space

How To: Finding a basis for the column space

Question. Given a $m \times n$ matrix A, find a basis for $\mathrm{Col}(A)$

How To: Finding a basis for the column space

Question. Given a $m \times n$ matrix A, find a basis for $\mathrm{Col}(A)$

We already know the columns of A span $\mathrm{Col}(A)$

How To: Finding a basis for the column space

Question. Given a $m \times n$ matrix A, find a basis for $\mathrm{Col}(A)$

We already know the columns of A span $\mathrm{Col}(A)$

So we also already know *some* subset of columns of A form a basis for $\mathrm{Col}(A)$

How To: Finding a basis for the column space

Question. Given a $m \times n$ matrix A, find a basis for $\text{Col}(A)$

We already know the columns of A span $\text{Col}(A)$

So we also already know *some* subset of columns of A form a basis for $\text{Col}(A)$

Which vectors should we choose?

Column Space and Reduced Echelon form

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3 \quad \mathbf{a}_4 \quad \mathbf{a}_5] \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Column Space and Reduced Echelon form

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3 \quad \mathbf{a}_4 \quad \mathbf{a}_5] \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The idea. What if we cover up the non-pivot columns?

Column Space and Reduced Echelon form

$$[\mathbf{a}_1 \quad \text{[redacted]} \quad \mathbf{a}_3 \quad \text{[redacted]}] \sim \begin{bmatrix} 1 & \text{[redacted]} & 0 & \text{[redacted]} \\ 0 & & 1 & \\ 0 & & 0 & \end{bmatrix}$$

The idea. What if we cover up the non-pivot columns?

Then we see $[\mathbf{a}_1 \quad \mathbf{a}_3]$ has 2 pivots

Column Space and Reduced Echelon form

$$\begin{bmatrix}\mathbf{a}_1 & \text{[redacted]} & \mathbf{a}_3 & \text{[redacted]} & \text{[redacted]}\end{bmatrix} \sim \begin{bmatrix}1 & \text{[redacted]} & 0 & \text{[redacted]} & \text{[redacted]} \\ 0 & & 1 & & \\ 0 & & 0 & & \end{bmatrix}$$

The idea. What if we cover up the non-pivot columns?

Then we see $\begin{bmatrix}\mathbf{a}_1 & \mathbf{a}_3\end{bmatrix}$ has 2 pivots

So the pivot columns are *linearly independent*

Column Space and Reduced Echelon form

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3 \quad \mathbf{a}_4 \quad \mathbf{a}_5] \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Column Space and Reduced Echelon form

$$\begin{array}{ccc} 2 & 1 & \\ \left[\mathbf{a}_1\right. & \mathbf{a}_2 & \left.\text{[redacted]}\right] \end{array} \sim \begin{array}{ccc} 2 & 1 & \\ \left[\begin{matrix} 1 \\ 0 \\ 0 \end{matrix}\right. & \begin{matrix} -2 \\ 0 \\ 0 \end{matrix} & \left.\text{[redacted]}\right] \end{array}$$

Observation. $[2\ 1\ 0\ 0\ 0]^T$ is a solution to the system $A\mathbf{x} = \mathbf{0}$

Column Space and Reduced Echelon form

$$\begin{array}{ccc} 2 & 1 & \\ \end{array}$$
$$\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \text{[redacted]} \end{bmatrix} \sim$$
$$\begin{array}{ccc} 2 & 1 & \\ \end{array}$$
$$\begin{bmatrix} 1 & -2 & \text{[redacted]} \\ 0 & 0 & \\ 0 & 0 & \end{bmatrix}$$

Observation. $[2\ 1\ 0\ 0\ 0]^T$ is a solution to the system $A\mathbf{x} = \mathbf{0}$

So $\mathbf{2a}_1 + \mathbf{a}_2 = \mathbf{0}$ and $\mathbf{a}_2 = (-2)\mathbf{a}_1$

Column Space and Reduced Echelon form

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3 \quad \mathbf{a}_4 \quad \mathbf{a}_5] \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Observation. $[2\ 1\ 0\ 0\ 0]^T$ is a solution to the system $A\mathbf{x} = \mathbf{0}$

So $\mathbf{2a}_1 + \mathbf{a}_2 = \mathbf{0}$ and $\mathbf{a}_2 = (-2)\mathbf{a}_1$

In general, every non-pivot column of A can be written as a linear combination pivots in front of it

Column Space and Reduced Echelon form

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3 \quad \mathbf{a}_4 \quad \mathbf{a}_5] \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Observation. $[2\ 1\ 0\ 0\ 0]^T$ is a solution to the system $A\mathbf{x} = \mathbf{0}$

So $\mathbf{2a}_1 + \mathbf{a}_2 = \mathbf{0}$ and $\mathbf{a}_2 = (-2)\mathbf{a}_1$

In general, every non-pivot column of A can be written as a linear combination pivots in front of it

This tells us that $\mathbf{a}_1$ and $\mathbf{a}_3$ *span* $\text{Col}(A)$

Column Space and Reduced Echelon form

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3 \quad \mathbf{a}_4 \quad \mathbf{a}_5] \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Column Space and Reduced Echelon form

$$\begin{bmatrix}\mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{a}_4 & \mathbf{a}_5\end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The takeaway. The pivot columns of A form a basis for $\mathrm{Col}(A)$

Column Space and Reduced Echelon form

$$\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{a}_4 & \mathbf{a}_5 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The takeaway. The pivot columns of A form a basis for $\mathsf{Col}(A)$

!! IMPORTANT !!

Choose the columns <u>of A</u>

($\mathbf{e}_1$ and $\mathbf{e}_2$ do not necessarily form a basis for $\mathsf{Col}(A)$)

How To: Finding a basis for the column space

Q. Given a $m \times n$ matrix A, find a basis for $\text{Col}(A)$

A.

1. Find the pivot columns in an echelon form of A
2. The associated columns <u>in A</u> form a basis for $\text{Col}(A)$

Question

$$A = \begin{bmatrix} 1 & -2 & 19 & 0 & -4 \\ 1 & 0 & 9 & 1 & 1 \\ 1 & -1 & 14 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 9 & 0 & 0 \\ 0 & 1 & -5 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

Find a bases for the column space and null space of A

Answer

$$A = \begin{bmatrix} 1 & -2 & 19 & 0 & -4 \\ 1 & 0 & 9 & 1 & 1 \\ 1 & -1 & 14 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 9 & 0 & 0 \\ 0 & 1 & -5 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

Summary

Subspaces define "tilted versions" of $\mathbb{R}^k$ in $\mathbb{R}^n$ (where $k \leq n$)

Bases are compact representation of subspaces as minimal spanning sets

Matrices have useful associated subspaces like the column space and null space